# AP Chemistry Unit 2: Compound Structure and Properties（去重版）

7 个课时共 115 题（答案版，各课时内沿用原编号）＋ Progress Check FRQ 1 大题
Progress Check MCQ 15 题经二次检查确认全部重复自课时题库，本册不再收录
各课时另有一份同内容纯题目版（未收录，按需可向老师索取）

## 2.1

**1.** The elements $C$ and $Se$ have the same electronegativity value, 2.55. Which of the following claims about the compound that forms from $C$ and $Se$ is most likely to be true?

(A) The carbon-to-selenium bond is unstable.

(B) The carbon-to-selenium bond is nonpolar covalent. ✓

(C) The compound has the empirical formula $CSe$.

(D) A molecule of the compound will have a partial negative charge on the carbon atom.

**Answer B**

Correct. The bonding in compounds between elements with the same (or only slightly different) electronegativity values will exhibit covalent bonding (that is, more or less equal sharing of electrons).

**2.** Two pure elements react to form a compound. One element is an alkali metal, $X$, and the other element is a halogen, $Z$. Which of the following is the most valid scientific claim that can be made about the compound?

(A) It has the formula $XZ_2$.

(B) It does not dissolve in water.

(C) It contains ionic bonds. ✓

(D) It contains covalent bonds.

**Answer C**

Correct. An alkali metal will readily lose an electron and a halogen will readily gain an electron to form an ionic compound with a $1{:}1$ ratio of $X^+$: $Z^-$ ions.

**3.** Which of the following scientific claims about the bond in the molecular compound $HF$ is most likely to be true?

(A) There is a partial negative charge on the $H$ atom.

(B) Electrons are shared equally between the $H$ and $F$ atoms.

(C) The bond is extremely weak.

(D) The bond is highly polar. ✓

**2.1**

**Answer D**

Correct. Fluorine is the most electronegative element, and the difference between the electronegativities of $\mathrm{H}$ and $\mathrm{F}$ is relatively large, resulting in a bond that is highly polar.

**4.**

| Element | Electronegativity |
|---|---|
| H | 2.1 |
| C | 2.5 |
| S | 2.5 |
| F | 4.0 |
| Cl | 3.0 |
| Si | 1.8 |

On the basis of the information above, which of the following arranges the binary compounds in order of increasing bond polarity?

(A) $CH_4 < SiCl_4 < SF_4$ ✓

(B) $CH_4 < SF_4 < SiCl_4$

(C) $SF_4 < CH_4 < SiCl_4$

(D) $SiCl_4 < SF_4 < CH_4$

**5.** Which of the following has the bonds arranged in order of decreasing polarity?

(A) H−F > N−F > F−F ✓

(B) H−I > H−Br > H−F

(C) O−N > O−S > O−Te

(D) Sb−I > Sb−Te > Sb−Cl

**6.** On the basis of electronegativity differences between atoms, which of the following scientific claims is the most accurate regarding the bonding in $\mathrm{Mg(NO_3)_2}(s)$ ?

**2.1**

(A) There is polar covalent bonding between $Mg$ atoms and $N$ atoms.

(B) There is polar covalent bonding between $Mg$ atoms and $O$ atoms.

(C) There is ionic bonding between $N^{5+}$ ions and $O^{2-}$ ions.

(D) There is ionic bonding between $Mg^{2+}$ ions and $NO_3^-$ ions. ✓

**Answer D**

Correct. There is ionic bonding in solid $Mg(NO_3)_2$ due to coulombic attractions between $Mg^{2+}$ ions and $NO_3^-$ ions. (There is some polar covalent bonding between $N$ and $O$ within the $NO_3^-$ ion.)

**7.** Which of the following bonds is most polar?

(A) F-F

(B) O-F

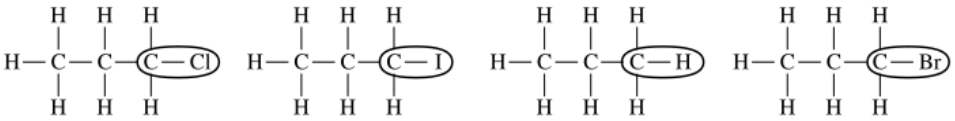
(C) O-I ✓

(D) O-Cl

**Answer C**

Correct. The polarity of bonds depends on the difference in the electronegativities of the atoms in the molecule; the greater the difference in electronegativity, the more polar the bond. Electronegativity generally decreases going down a column and from right to left across a row. Of the pairs given in the options, oxygen and iodine are farthest apart, have the greatest difference in electronegativity, and therefore form bonds that are the most polar.

**8.**




Which of the following bonds represented in the diagram above is the most polar?

(A) $C-Cl$ ✓

(B) $C-I$

(C) $C-H$

(D) $C-Br$

**2.1**

**Answer A**

Correct. The $C-Cl$ bond is the most polar of the bonds in the responses because $Cl$ is the most electronegative of $Cl$, $I$, $H$ and $Br$. This can be inferred given that $C$ and $H$ have similar electronegativities and the periodic trend is for electronegativity to decrease from the top to the bottom of a column.

**9.** Which of the following compounds contains both ionic and covalent bonds?

(A) $C_2H_5OH$

(B) $MgF_2$

(C) $H_2S$

(D) $NH_4Cl$ ✓

**10.** Which of the following claims about a binary compound composed of elements with the same electronegativity is most likely to be true?

(A) The compound has properties similar to those of both elements.

(B) The bonding in the compound is nonpolar covalent. ✓

(C) The boiling point of the compound is above $1000^\circ C$.

(D) The compound contains strong ionic bonds.

**Answer B**

Correct. Elements with the same electronegativity are more likely to share electrons and form a nonpolar covalent bond.

**11.** Which of the following substances has bonds with the greatest ionic character?

(A) $CCl_4$

(B) $CuCl_2$

(C) $CaCl_2$ ✓

(D) $NCl_3$

**2.1**

**Answer C**

Correct. The degree of ionic character in a chemical bond is a function of the difference in electronegativities of the two atoms that form the bond. Among C, Cu, Ca, and N, the element with the lowest electronegativity is Ca as, it is to the left of the other elements on the periodic table. Thus, the difference between the electronegativity of Ca and the electronegativity of Cl is the largest difference, and the Ca – Cl bond has the greatest ionic character.

**12.**

$$PCl_3, Cl_2, CaCl_2$$

Which of the following correctly ranks the substances listed above in order of increasing ionic character in their bonding?

(A) $Cl_2$, $PCl_3$, $CaCl_2$ ✓

(B) $Cl_2$, $CaCl_2$, $PCl_3$

(C) $CaCl_2$, $PCl_3$, $Cl_2$

(D) $CaCl_2$, $Cl_2$, $PCl_3$

**Answer A**

Correct. The ionic character of a bond increases with increasing differences between the electronegativities of the bonded elements. The bond in $Cl_2$ is the least ionic because the two bonded atoms are identical. The bonds in $CaCl_2$ are the most ionic because the bonded atoms (Ca and Cl) have the greatest difference in their electronegativities, which can be inferred from their relative positions in the periodic table. Ca and Cl are on opposite sides of the periodic table, so the difference in their electronegativities is relatively large. The bonds in $CaCl_2$ are more ionic than those in $PCl_3$ because the difference in electronegativities between Ca and Cl is greater than that between P and Cl.

**13.** An atom of chlorine and an atom of which of the following elements will form a bond with the most ionic character?

(A) Germanium

(B) Carbon

(C) Copper

(D) Potassium ✓

**2.1**

**Answer D**

Correct. The percent ionic character in a bond between two atoms increases with the difference in electronegativity between the bonded atoms. The electronegativity of elements varies with their position in the periodic table, increasing from left to right across a period and from bottom to top in a group. The greatest difference in electronegativity will be between chlorine and potassium because potassium is farther from chlorine on the periodic table than germanium, carbon, or copper are. Thus, the potassium-chlorine bond (K-Cl) will have the most ionic character.

**14.** Which of the following claims about a binary compound in which the bonding is ionic is most likely to be scientifically valid?

(A) Both elements in the compound are metals.

(B) The atomic masses of the elements in the compound are relatively small.

(C) There is equal sharing of electrons between the atoms of the elements in the compound.

(D) The electronegativity difference between the elements in the compound is relatively large. ✓

**Answer D**

Correct. A relatively large difference between the electronegativities of the two elements indicates that one element has a tendency to form cations and the other has a tendency to form anions.

**15.** Of the following compounds, which is the most ionic?

(A) $SiCl_4$

(B) BrCl

(C) $PCl_3$

(D) $CaCl_2$ ✓

**16.** Of the following single bonds, which is the LEAST polar?

(A) N—H

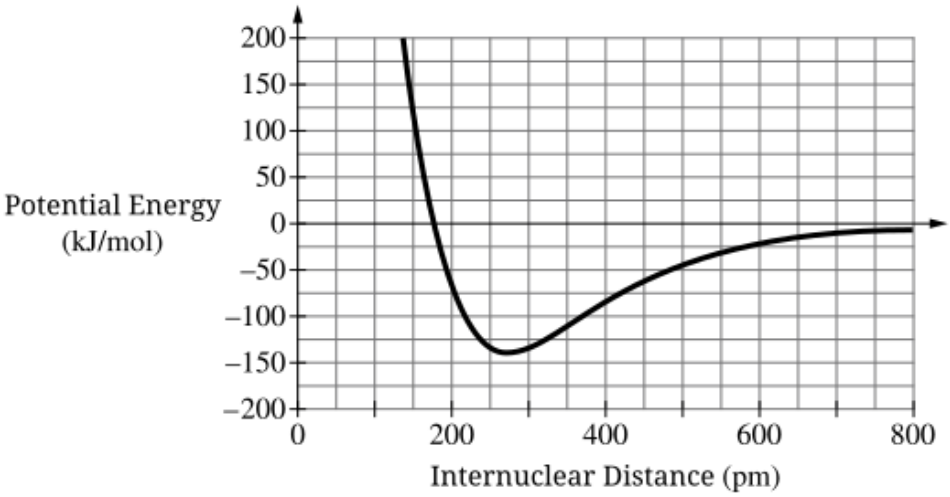
(B) H—F

(C) O—F ✓

(D) I—F

## 2.2

The graph shows the potential energy of two atoms as a function of internuclear distance.




**1.** What is the approximate bond energy?

(A) $140\ \text{kJ/mol}$ ✓

(B) $170\ \text{kJ/mol}$

(C) $200\ \text{kJ/mol}$

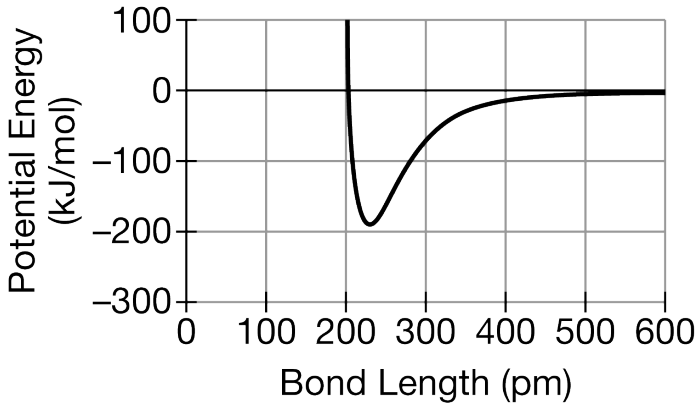
(D) $270\ \text{kJ/mol}$

**Answer A**

Correct. The minimum in the potential energy curve is $-140\ \text{kJ/mol}$. Bond energy is the energy required to break a bond and has the corresponding positive value, $140\ \text{kJ/mol}$.

**2.2**

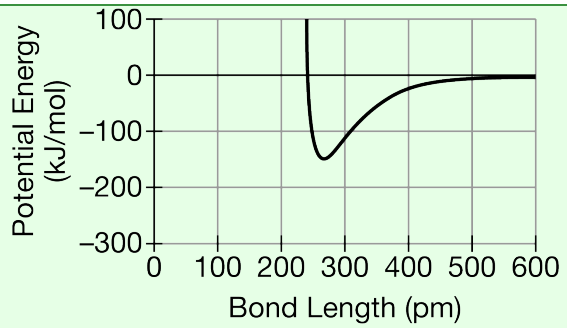
The following graph shows potential energy versus bond length for a $Br_2$ molecule.



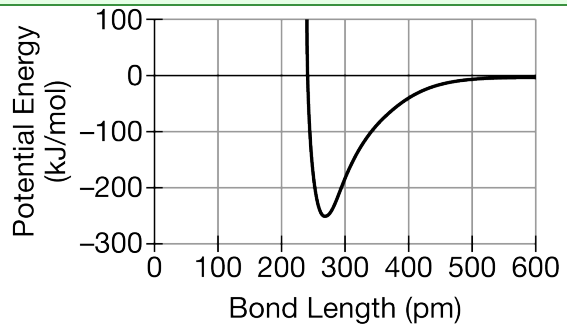

**2.** Which of the following graphs shows potential energy versus bond length for an $I_2$ molecule?

**2.2**

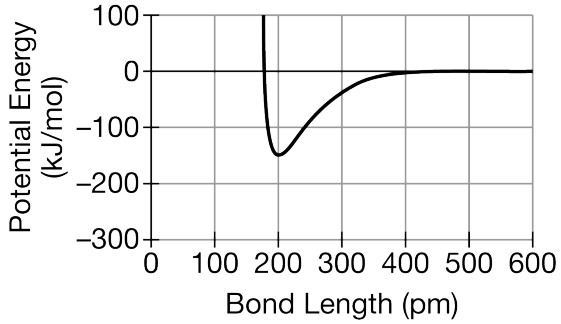
(A)



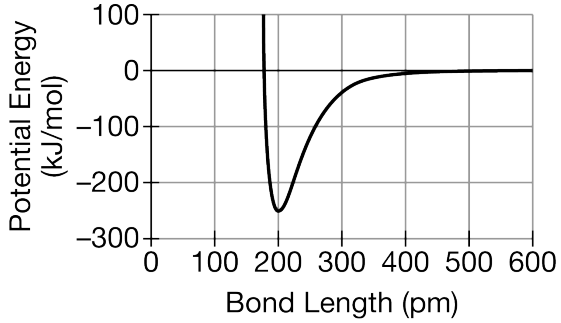

(B)




(C)




(D)



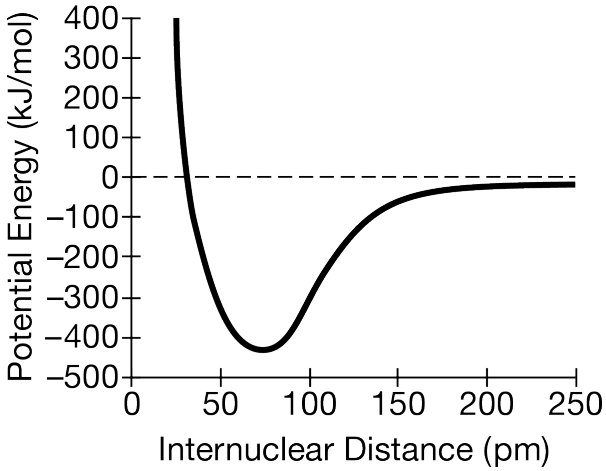

**Answer A**

Correct. $\mathrm{Br}$ and $\mathrm{I}$ are in the same group of the periodic table, but $\mathrm{I}$ has more occupied shells than $\mathrm{Br}$ does, so the $\mathrm{I}$ atom has a larger radius than the $\mathrm{Br}$ atom and its valence electrons are farther from the nucleus. This means that the $\mathrm{I}_2$ molecule has a longer bond length (the lowest point on the curve is farther to the right) than the $\mathrm{Br}_2$ molecule, and $\mathrm{I}_2$ has a bond energy with a smaller magnitude (the

**2.2**

lowest point on the curve is not as far down) than $Br_2$ does.

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**3.** Which of the following can be inferred from the diagram above that shows the dependence of potential energy on the internuclear distance between two atoms?

(A) The atoms form a bond with a bond length of $25\ \text{pm}$.

(B) The atoms form a bond with a bond length of $75\ \text{pm}$. ✓

(C) The net force between the atoms is attractive at $25\ \text{pm}$.

(D) The net force between the atoms is attractive at $75\ \text{pm}$.

**Answer B**

Correct. The minimum potential energy occurs at an internuclear distance of $75\ \text{pm}$, which corresponds to the length of the stable bond that forms between the two atoms. If the atoms were any closer to each other, the net force would be repulsive. Likewise, if the atoms were farther from each other, the net force would be attractive. At $75\ \text{pm}$, the net force between the two atoms is zero.

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**4.** Which of the following compounds has the strongest Coulombic attractions between its ions in the solid state?

(A) $MgO$ ✓

(B) $CaO$

(C) $MgCl_2$

(D) $CaCl_2$

**2.2**

**Answer A**

Correct. The Coulombic force is directly proportional to $\frac{q_1 q_2}{r^2}$, where $q_1$ and $q_2$ are the charges of the ions and $r$ is the distance between the ions. Ionic compounds that contain the smallest ions and ions with the greatest charges therefore have the strongest Coulombic attractions between ions. The charge on the cation is $+2$ in all responses. $\mathrm{MgO}$ and $\mathrm{CaO}$ have a $-2$ charge on the anion, whereas $\mathrm{MgCl_2}$ and $\mathrm{CaCl_2}$ have a $-1$ charge on the anion. The $\mathrm{Mg^{2+}}$ ion has one less occupied electron shell and thus is smaller than the $\mathrm{Ca^{2+}}$ ion, so $\mathrm{MgO}$ has the strongest Coulombic attractions among the choices given.

**5.**

| Compound | Melting Point ($^\circ\mathrm{C}$) |
|---|---|
| $\mathrm{MgO}$ | 2852 |
| $\mathrm{NaF}$ | 993 |

The table above shows the melting points of $\mathrm{MgO}(s)$ and $\mathrm{NaF}(s)$. Which of the following best helps explain why the melting point of $\mathrm{MgO}(s)$ is much higher than that of $\mathrm{NaF}(s)$ ?

(A) The mass of $\mathrm{F^-}$ ions is greater than that of $\mathrm{O^{2-}}$ ions.

(B) The mass of $\mathrm{Mg^{2+}}$ ions is greater than that of $\mathrm{Na^+}$ ions.

(C) The difference between the electronegativity values of $\mathrm{Mg}$ and $\mathrm{O}$ is smaller than the difference between the values of $\mathrm{Na}$ and $\mathrm{F}$.

(D) The charges of $\mathrm{Mg^{2+}}$ and $\mathrm{O^{2-}}$ ions are greater than those of $\mathrm{Na^+}$ and $\mathrm{F^-}$ ions. ✓

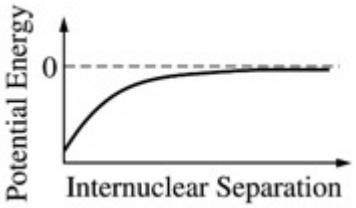
**Answer D**

Correct. The melting point of an ionic lattice tends to be higher the stronger the force of attraction between the cation and the anion. According to Coulomb's law, the greater the charges, the stronger the force of attraction and the higher the melting point. $\mathrm{MgO}$ involves a $+2$ cation and a $-2$ anion, whereas $\mathrm{NaF}$ involves a $+1$ cation and a $-1$ anion. Hence, $\mathrm{MgO}$ has the higher melting point.

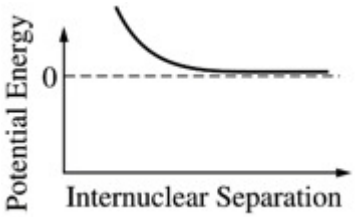
**6.** Which of the following graphs correctly shows the relationship between potential energy and internuclear separation for two hydrogen atoms?

**2.2**

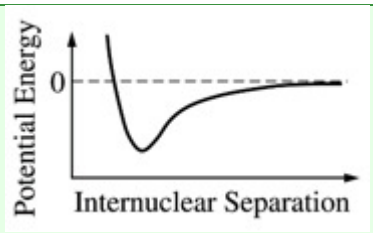
(A)


Potential Energy
0
Internuclear Separation

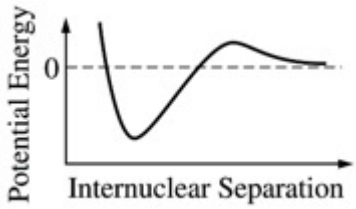
(B)


Potential Energy
0
Internuclear Separation

(C) ✓

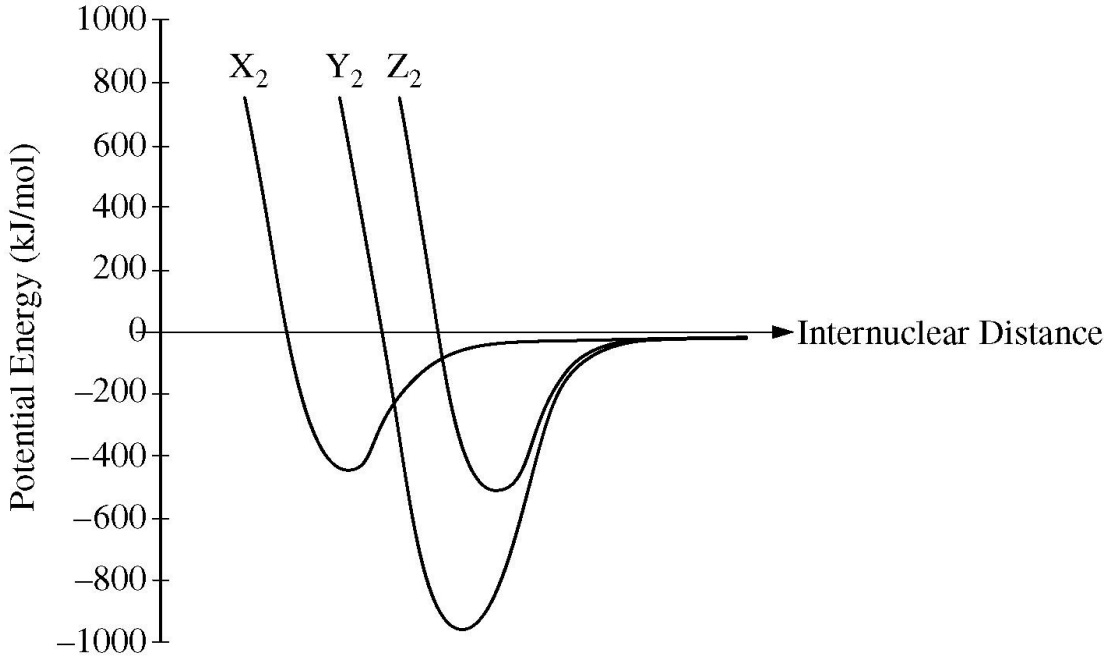

Potential Energy
0
Internuclear Separation

(D)


Potential Energy
0
Internuclear Separation

**2.2**

**7.**




The potential energy as a function of internuclear distance for three diatomic molecules, $X_2$, $Y_2$, and $Z_2$, is shown in the graph above. Based on the data in the graph, which of the following correctly identifies the diatomic molecules, $X_2$, $Y_2$, and $Z_2$?

(A) ✓

| **$X_2$** | **$Y_2$** | **$Z_2$** |
|---|---|---|
| $H_2$ | $N_2$ | $O_2$ |

(B)

| **$X_2$** | **$Y_2$** | **$Z_2$** |
|---|---|---|
| $H_2$ | $O_2$ | $N_2$ |

(C)

| **$X_2$** | **$Y_2$** | **$Z_2$** |
|---|---|---|
| $N_2$ | $O_2$ | $H_2$ |

(D)

| **$X_2$** | **$Y_2$** | **$Z_2$** |
|---|---|---|
| $O_2$ | $H_2$ | $N_2$ |

**2.2**

**8.** The lattice energy of a salt is related to the energy required to separate the ions. For which of the following pairs of ions is the energy that is required to separate the ions largest? (Assume that the distance between the ions in each pair is equal to the sum of the ionic radii.)

(A) $Na^+(g)$ and $Cl^-(g)$

(B) $Cs^+(g)$ and $Br^-(g)$

(C) $Mg^{2+}(g)$ and $O^{2-}(g)$ ✓

(D) $Ca^{2+}(g)$ and $O^{2-}(g)$

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Lattice enthalpy can be defined as the $\Delta H^\circ$ associated with the separation of a solid crystal into gaseous ions. Lattice enthalpies for four compounds are given in the table.

| Compound | Lattice Enthalpy $(kJ/mol)$ |
|---|---|
| $LiF$ | 1030 |
| $LiCl$ | 834 |
| $MgCl_2$ | 2510 |
| $SrCl_2$ | 2127 |

**9.** Which of the following statements about lattice enthalpy is consistent with the information in the table?

(A) It increases with the increasing radii of the ions in the compound.

(B) It increases with the decreasing radii of the ions in the compound. ✓

(C) It increases with the decreasing charge of the ions.

(D) It increases with the decreasing electronegativity of the atoms.

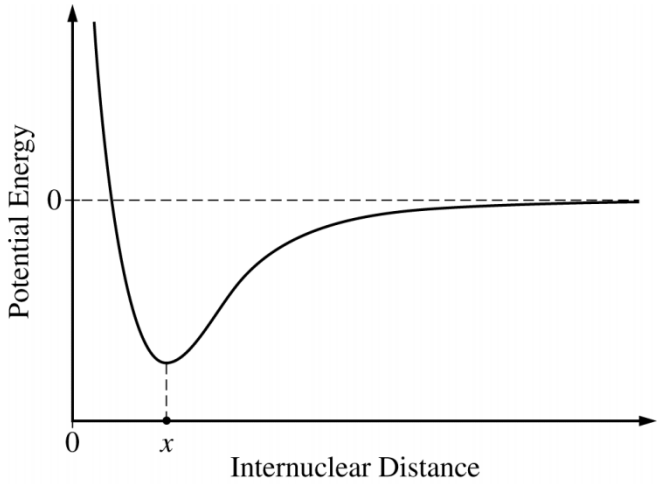
**Answer B**

Correct. $F^-$ has fewer occupied electron shells and is a smaller ion than $Cl^-$ is, and $LiF$ has a larger lattice enthalpy than $LiCl$ does. Similarly, $Mg^{2+}$ is a smaller ion than $Sr^{2+}$ is, and $MgCl_2$ has a larger lattice enthalpy than $SrCl_2$ does. Therefore, this statement is consistent with the data in the table.

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**2.2**

**10.**




The potential energy of a system of two atoms as a function of their internuclear distance is shown in the diagram above. Which of the following is true regarding the forces between the atoms when their internuclear distance is $x$?

(A) The attractive and repulsive forces are balanced, so the atoms will maintain an average internuclear distance $x$. ✓

(B) There is a net repulsive force pushing the atoms apart, so the atoms will move further apart.

(C) There is a net attractive force pulling the atoms together, so the atoms will move closer together.

(D) It cannot be determined whether the forces between atoms are balanced, attractive, or repulsive, because the diagram shows only the potential energy

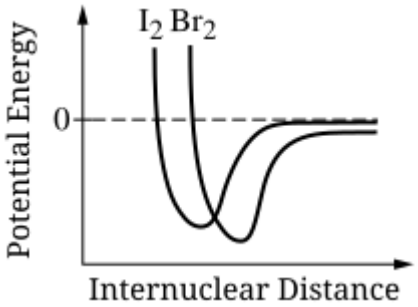
**11.**

| Molecule | Bond Dissociation Energy $(\text{kJ / mol})$ |
|---|---|
| $\text{Br}_2$ | 193 |
| $\text{I}_2$ | 151 |

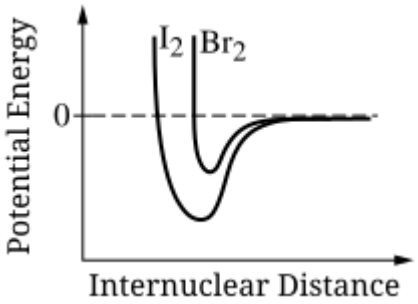
The table above shows the bond dissociation energies for $\text{Br}_2$ and $\text{I}_2$ at $298\ \text{K}$. Which of the following correctly shows the potential energy as a function of internuclear distance for two bromine atoms and two iodine atoms?

**2.2**

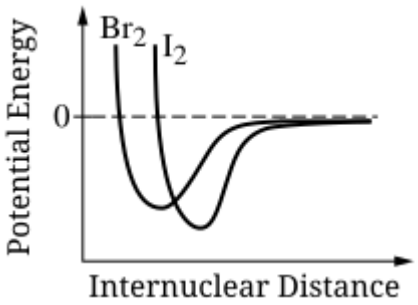
(A)



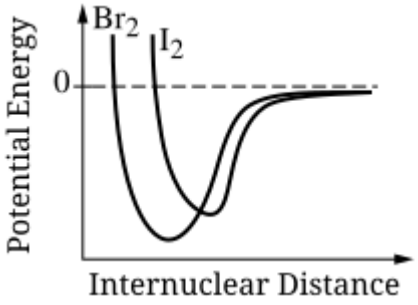

(B)




(C)




(D)

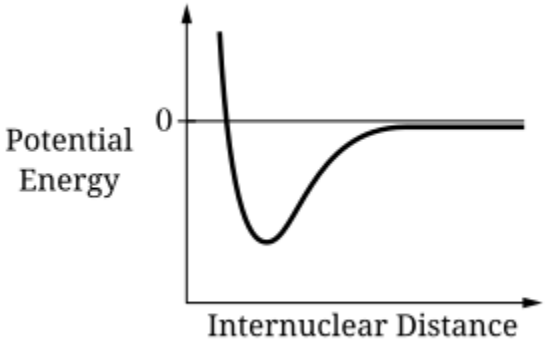



**Answer D**

Correct. The bond dissociation energy for $Br_2$ is given to be greater than that for $I_2$. In addition, $Br$ is smaller than $I$, so the equilibrium internuclear distance for $Br_2$ should be less than that for $I_2$.

**2.2**

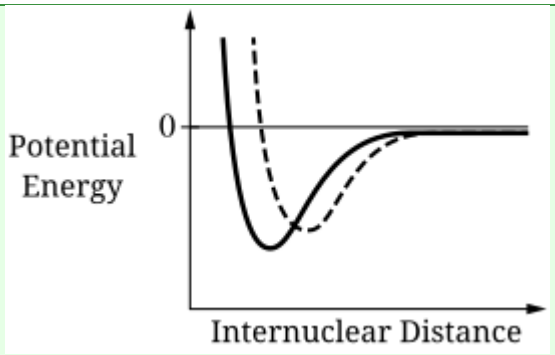
**12.**



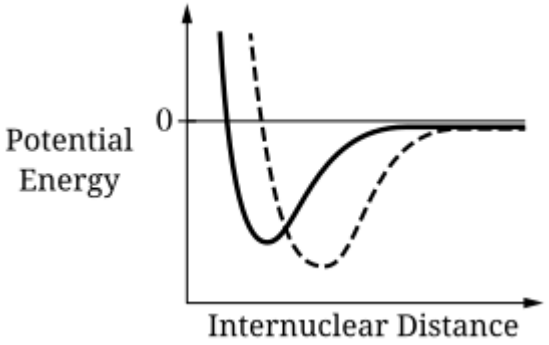

The graph of potential energy versus internuclear distance for the $HCl$ molecule is shown. Which of the following graphs includes the plot for the $HBr$ molecule, shown as a dashed line?

**2.2**

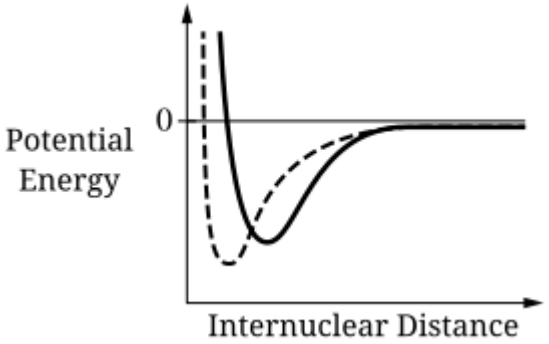
(A)



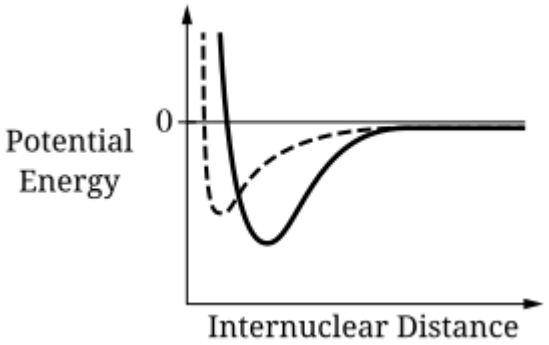

(B)




(C)




(D)




**Answer A**

Correct. $HBr$ will have an equilibrium internuclear distance that is greater than that for $HCl$. In addition, $HBr$ will have a smaller bond strength (depth of well) than $HCl$. The graph satisfies both these

**2.2**

statements.

---

Gaseous hydrogen and gaseous chlorine can react to form hydrogen chloride gas, as represented by the following equation.

$$H_2(g) + Cl_2(g) \rightarrow 2\ HCl(g)$$

**13.** Which of the following correctly ranks the internuclear distances for the $H_2$, $Cl_2$, and $HCl$ molecules from shortest to longest?

(A) $H_2 < Cl_2 < HCl$

(B) $H_2 < HCl < Cl_2$ ✓

(C) $Cl_2 < HCl < H_2$

(D) $Cl_2 < H_2 < HCl$

**Answer B**

Correct. The atomic radius of a $Cl$ atom is much greater than that of an $H$ atom because $Cl$ has more shells of electrons than $H$ does. The electron configuration of an $H$ atom is $1s^1$, and the electron configuration of a $Cl$ atom is $1s^2 2s^2 2p^6 3s^2 3p^5$. Because of the respective atomic sizes of $H$ and $Cl$, $H_2$ should have the shortest bond distance and $Cl_2$ should have the longest bond distance of the three molecules.

---

**14.** The melting point of MgO is higher than that of NaF. Explanations for this observation include which of the following?

I. $Mg^{2+}$ is more positively charged than $Na^{+}$.

II. $O^{2-}$ is more negatively charged than $F^{-}$.

III. The $O^{2-}$ ion is smaller than the $F^{-}$ ion.

(A) I and II only ✓

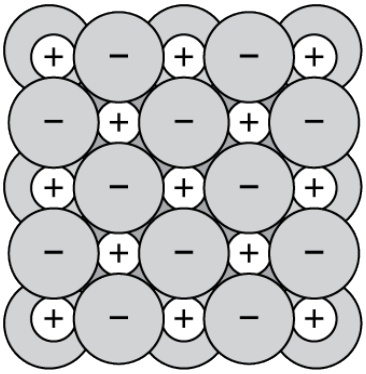
(B) I and III only

(C) II and III only

(D) I, II, and III

## 2.3

**1.**



Which of the following correctly indicates whether the solid represented by the particulate model shown above conducts electricity and explains why or why not?

(A) It conducts electricity because it is made of positive and negative ions.

(B) It conducts electricity because it is made of particles of different sizes.

(C) It does not conduct electricity because its ions cannot move freely within the solid. ✓

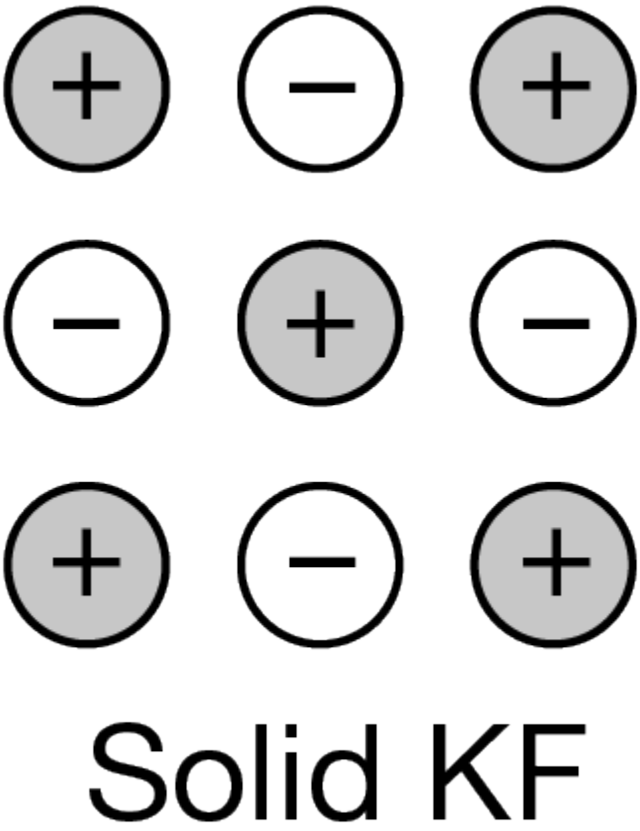
(D) It does not conduct electricity because there are small spaces between the particles.

**Answer C**

Correct. In the solid state, the ions are held by strong coulombic forces in fixed positions within the lattice; thus they cannot move or migrate through the lattice.

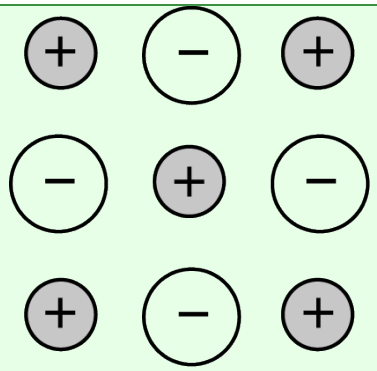
**2.3**

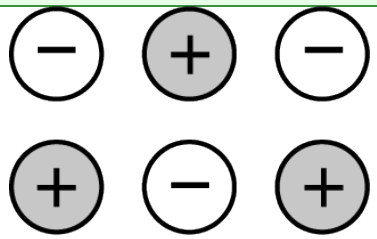
**2.**

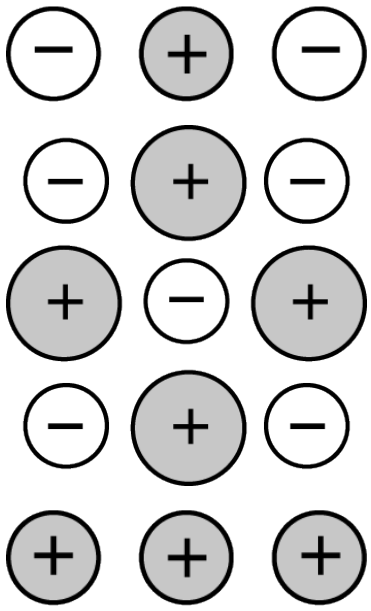


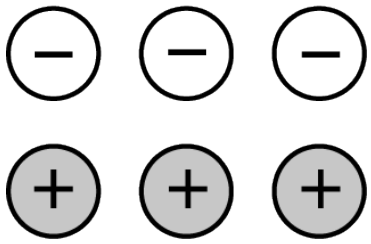

The particle-level diagram above represents the structure of solid $\mathrm{KF}$. Although the molar mass of $\mathrm{KCl}$ is greater than that of $\mathrm{KF}$, the density of $\mathrm{KCl}$ is actually less than that of $\mathrm{KF}$. Which of the following representations of the structure of $\mathrm{KCl}$ best helps to explain this phenomenon?

**2.3**

(A) 

(B) 

(C) 

(D) 

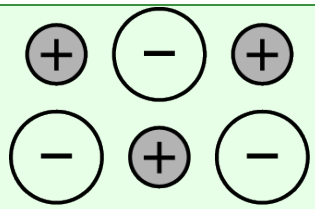
**Answer A**

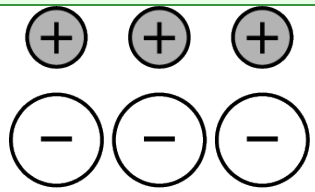
Correct. The diagram shows that the chloride ions are larger than the fluoride ions, which causes the

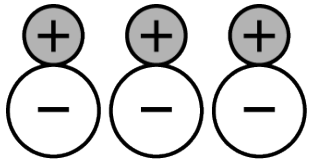
**2.3**

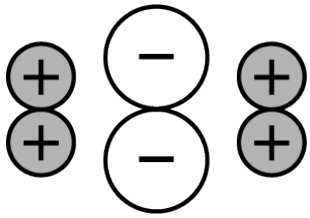
interionic distances in the solid to increase to the point where the effect of the larger spacing among ions in $KCl$ is greater than the effect of the larger mass of the chlorine on the density.

**3.** The particles in solid $KI$, a stable ionic compound, are arranged to maximize coulombic attractions while minimizing coulombic repulsions among the particles. Which of the following diagrams best represents the structure of solid $KI$?

(A) 

(B) 

(C) 

(D) 

**Answer A**

Correct. This diagram shows both types of charged ions being surrounded by ions of the opposite charge, thus maximizing attraction and minimizing repulsion.

**2.3**

**4.**

| Compound | Lattice Energy $(\mathrm{kJ/mol})$ |
|---|---|
| $\mathrm{NaF}$ | 930 |
| $\mathrm{MgF_2}$ | 2978 |

The energy required to dissociate an ionic solid into gaseous ions (lattice energy) for the compounds $\mathrm{NaF}$ and $\mathrm{MgF_2}$ is shown in the table above. On the basis of Coulomb's law, which of the following best helps to explain the large difference between the lattice energies of $\mathrm{NaF}$ and $\mathrm{MgF_2}$ ?

(A) The solubility of $\mathrm{MgF_2}$ is less than that of $\mathrm{NaF}$.

(B) The electronegativity of $\mathrm{Mg}$ is greater than that of $\mathrm{Na}$.

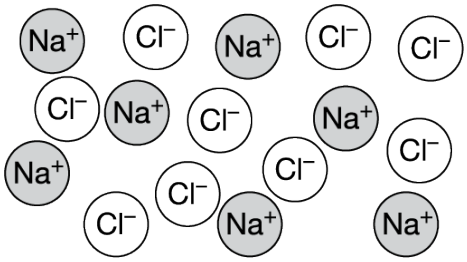
(C) The mass of the $\mathrm{Mg}$ cation is greater than that of the $\mathrm{Na}$ cation.

(D) The charge of the $\mathrm{Mg}$ cation is larger than that of the $\mathrm{Na}$ cation. ✓

**Answer D**

Correct. The coulombic attractions between the $\mathrm{Mg}$ cation and $\mathrm{F}$ anions are stronger because of the $+2$ charge on $\mathrm{Mg}$ as opposed to the $+1$ charge on $\mathrm{Na}$. Stronger interionic attractions lead to a higher lattice energy.

**5.**




Molten (liquid) $\mathrm{NaCl}$ is represented by the particulate diagram shown above. Which of the following indicates whether $\mathrm{NaCl}(l)$ conducts electricity and best explains why or why not?

(A) It conducts electricity because $\mathrm{Na}$ is a metal.

(B) It conducts electricity because ions are free to move. ✓

(C) It does not conduct electricity because $\mathrm{Cl}$ is a nonmetal.

(D) It does not conduct electricity because there are no free electrons.

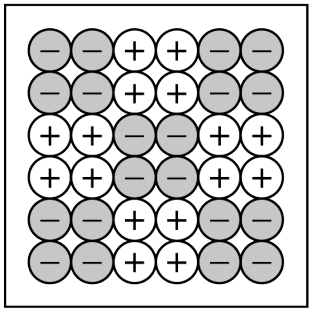
**2.3**

**Answer B**

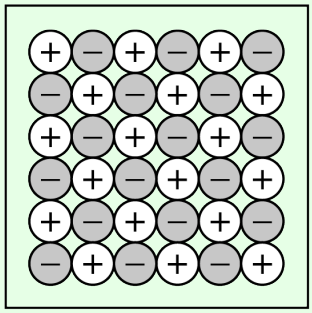
Correct. For a substance to conduct electricity, it must contain charged particles that are mobile. In molten salts such as $NaCl$, those charged particles are ions. (Metals conduct electricity via mobile electrons within a lattice of metal atoms.)

**6.** Of the following diagrams, which best represents the structure of solid $KF$ ?

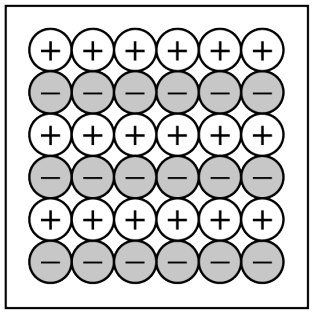
(A)



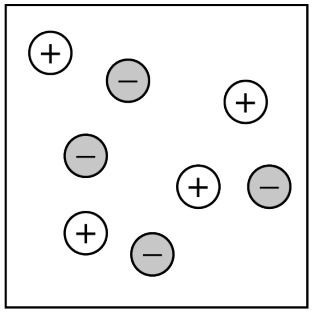
(B)



(C)



(D)



**2.3**

**Answer B**

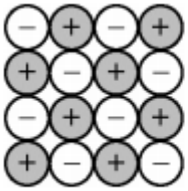
Correct. This diagram shows ions as being tightly packed surrounded by ions of the opposite charge, which is true in a crystal of an ionic compound such as $KF$. This arrangement represents the lowest energy state attainable by the constituent ions.

**7.** Which of the following diagrams best illustrates how a displacement in an ionic crystal results in cleavage and brittleness?

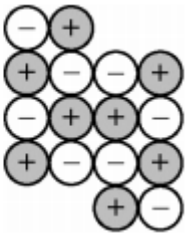
**2.3**

(A) ✓

Before Displacement

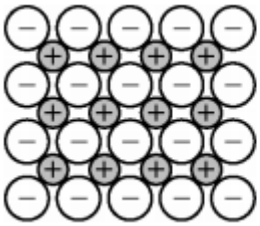


After Displacement

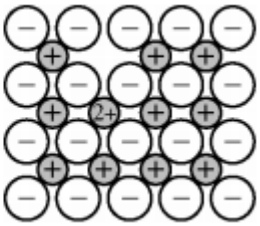


(B)

Before Displacement

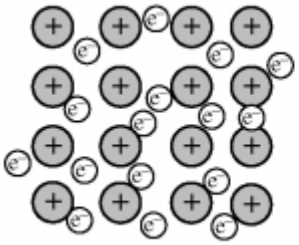


After Displacement

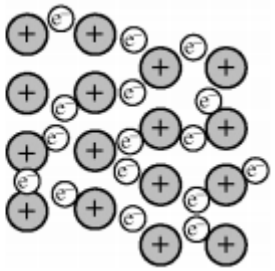


(C)

Before Displacement

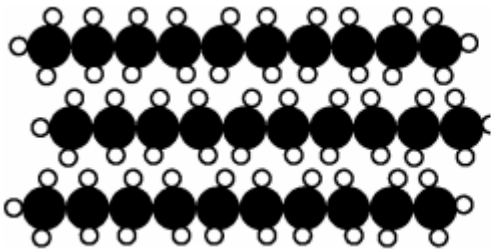


After Displacement

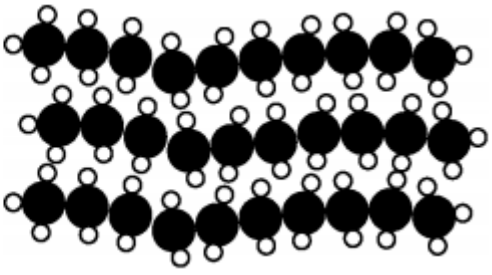


(D)

Before Displacement

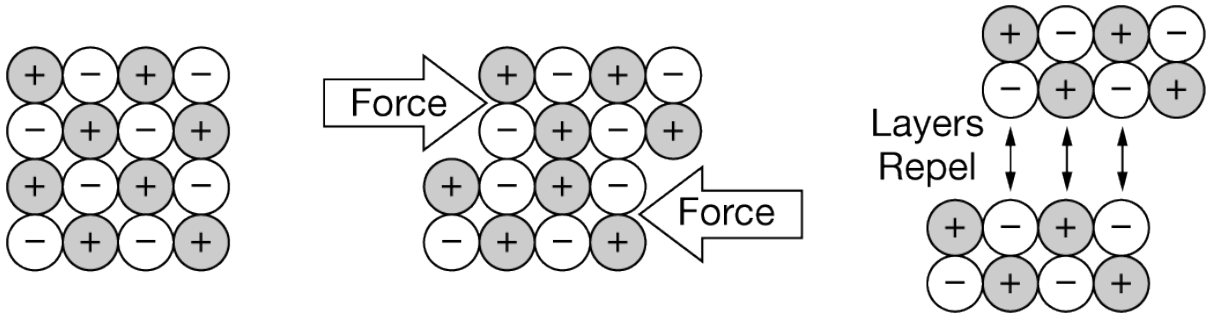


After Displacement



**2.3**

**8.**




The particulate-level diagram shown above best helps to explain which of the following properties of ionic solids?

(A) Density

(B) Brittleness ✓

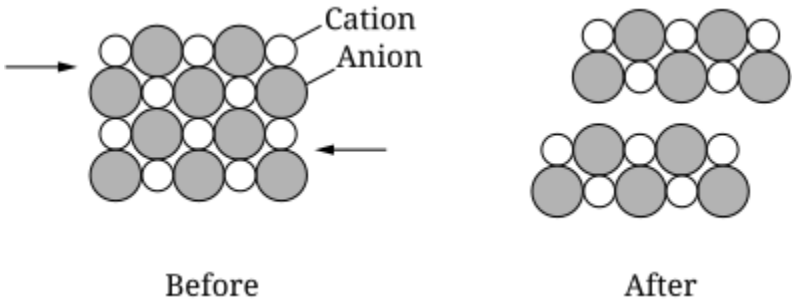
(C) Malleability

(D) Conductivity

**Answer B**

Correct. The brittleness of ionic solids is due to the internal repulsion of ion layers within the crystal lattice that occurs when the layers are displaced by an outside force parallel to the layers and they move relative to adjacent layers.

**9.**




When a stress is applied to an ionic crystal, the planes of the ions slide and separate due to the alignment of like charges, as shown in the diagram above. Which of the following characteristics of the crystal is represented by the observation?

(A) Luster

(B) Brittleness ✓

(C) High melting point

(D) Solubility in water

**2.3**

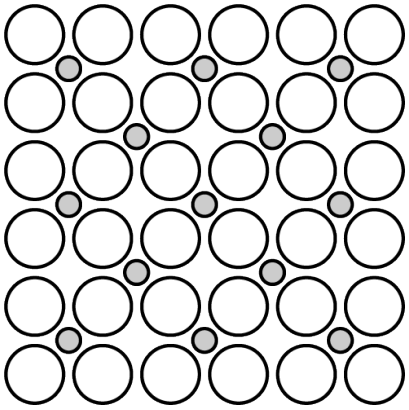
**Answer B**

Correct. Brittleness is the tendency of a material to break when pressure is applied.

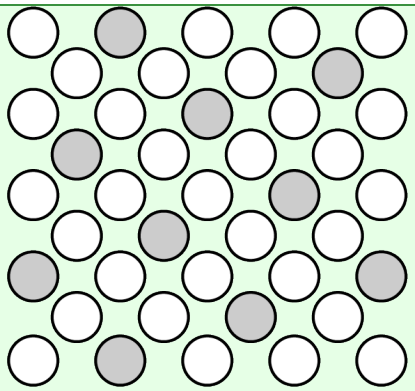
## 2.4

**1.** Copper atoms and zinc atoms have the same atomic radius, 135 picometers. Based on this information, which of the following diagrams best represents an alloy containing only copper and zinc atoms?

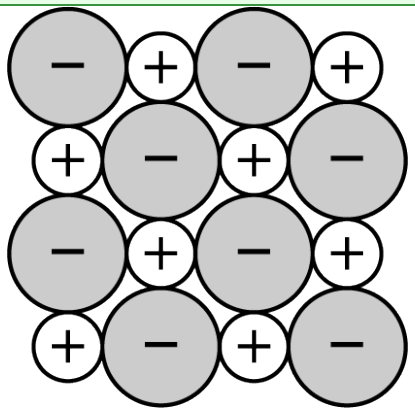
(A)



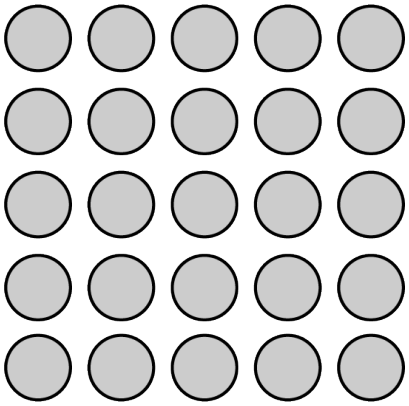
(B) ✓



(C)



(D)

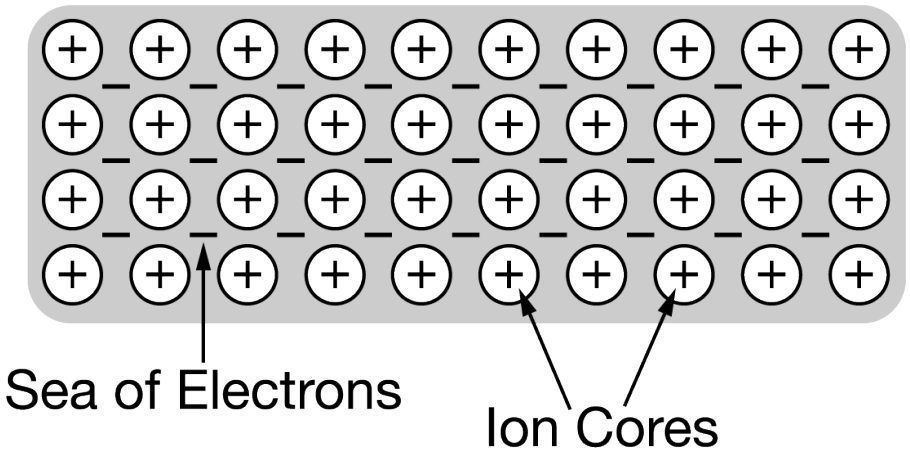


**2.4**

**Answer B**

Correct. Being the same size, copper atoms and zinc atoms would form a substitutional alloy in which the atoms substitute for one another in a lattice of positive cores immersed in a sea of electrons.

**2.**




A particle-level diagram of a metallic element is shown above. Typically, metals are both malleable and ductile. The best explanation for these properties is that the electrons involved in bonding among metal atoms are

(A) unequally shared and form nondirectional bonds

(B) unequally shared and form highly directional bonds

(C) equally shared and form nondirectional bonds ✓

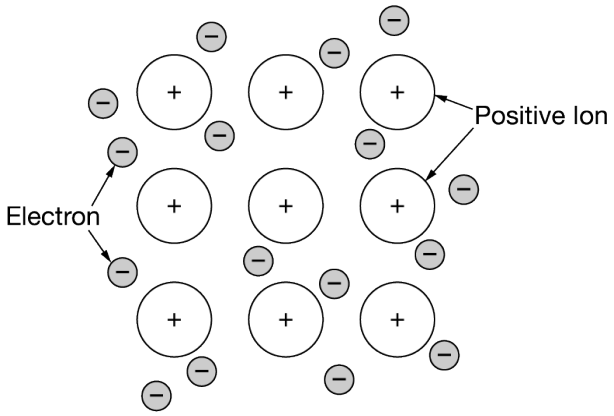
(D) equally shared and form highly directional bonds

**Answer C**

Correct. In metals, electrons are free to move as described by the electron sea model. Electrons are shared equally among atoms, and the structure is based on simple close-packed structures that contain no fixed directional bonds.

**2.4**

**3.**




Which of the following indicates whether the solid substance represented by the particulate diagram shown above conducts electricity and explains why or why not?

(A) It conducts electricity because it is made of positive and negative particles.

(B) It conducts electricity because electrons are free to move through the substance. ✓

(C) It does not conduct electricity because electrons are strongly attracted to specific positive particles.

(D) It does not conduct electricity because the positive particles are not free to move through the substance.

**Answer B**

Correct. The diagram shows the structure of a metallic solid, which consists of a lattice of positive ions immersed in a sea of electrons in which electrons can move freely.

**4.**

| Element | Metallic Radius (pm) | Melting Point (°C) | Common Oxidation State |
|---|---|---|---|
| Au | 144 | 1064 | 1+, 3+ |
| Cu | 128 | 1085 | 1+, 2+ |
| Ag | 144 | 961 | 1+ |

To make Au stronger and harder, it is often alloyed with other metals, such as Cu and Ag. Consider two alloys, one of Au and Cu and one of Au and Ag, each with the same mole fraction of Au. If the Au/Cu alloy is harder than the Au/Ag alloy, then which of the following is the best explanation based on the information in the table above?

(A) Cu has two common oxidation states, but Ag has only one.

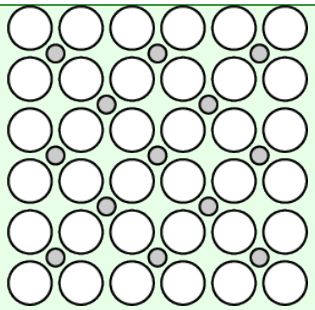
(B) Cu has a higher melting point than Au has, but Ag has a lower melting point than Au has.

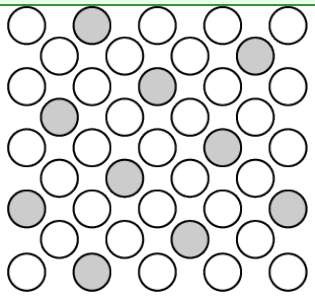
(C) Cu atoms are smaller than Ag atoms, thus they interfere more with the displacement of atoms in the alloy. ✓

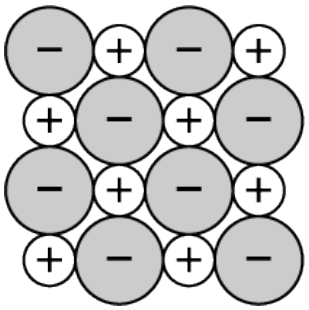
(D) Cu atoms are less polarizable than are Au or Ag atoms, thus Cu has weaker interparticle forces.

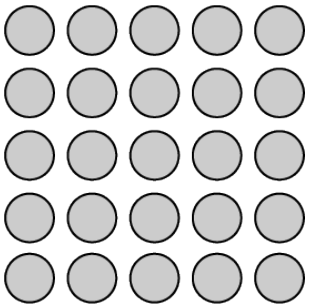
**2.4**

**5.** Which of the following particulate-level diagrams best represents an interstitial alloy?

(A) 

(B) 

(C) 

(D) 

**Answer A**

Correct. In an interstitial alloy, atoms of one of the constituents are significantly smaller than the atoms of the other constituent and occupy the smaller spaces (interstices) among the larger atoms.

**2.4**

Atomic radii and electronegativity values for some elements are given in the table.

| Element | Radius | Electronegativity |
|---|---|---|
| Ag | 165 pm | 1.93 |
| Au | 174 pm | 2.54 |
| Cu | 145 pm | 1.90 |
| N | 56 pm | 3.04 |
| Ni | 149 pm | 1.91 |
| W | 193 pm | 2.36 |

**6.** Which of the following names a pair of elements that would likely form a substitutional alloy and explains why?

(A) Ni and Cu, because Ni atoms would fit in the spaces between Cu atoms in the solid lattice.

(B) Ni and Cu, because Ni atoms would take the place of Cu atoms in the solid lattice. ✓

(C) W and N, because N atoms would fit in the spaces between W atoms in the solid lattice.

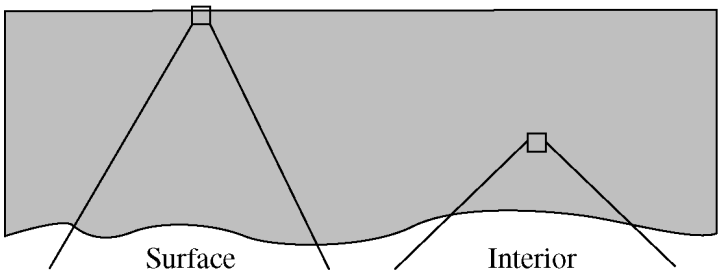
(D) W and N, because N atoms would take the place of W atoms in the solid lattice.

**Answer B**

Correct. Substitutional alloys form between elements with atoms of comparable radius, where atoms of one element substitute for atoms of another element in the solid lattice. Ni and Cu have similar radii.

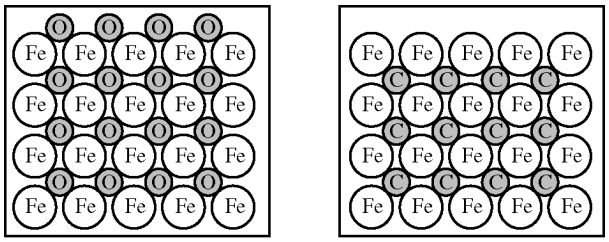
**2.4**

**7.** Steel is an alloy consisting of Fe with a small amount of C. Elemental Cr can be added to steel to make the steel less likely to rust; Cr atoms react with oxygen in the air to form a nonreactive layer of chromium oxide on the surface of the steel, preventing the oxidation of underlying Fe atoms. A sample of steel-chromium alloy contains 15 percent Cr by mass. Which of the following diagrams best shows a particle-level view of a surface section and an interior section of the alloy represented below at the left? (The atomic radii of the atoms involved are given in the table below at the right.)

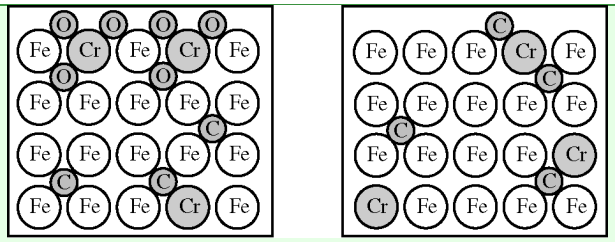



| Element | Molar Mass (g/mol) | Atomic Radius (pm) |
|---|---|---|
| Fe | 55.85 | 125 |
| Cr | 52.00 | 127 |
| C | 12.01 | 77 |
| O | 16.00 | 73 |

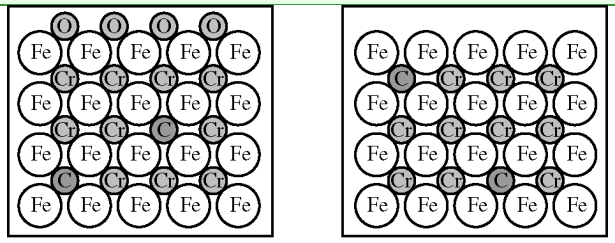
(A)



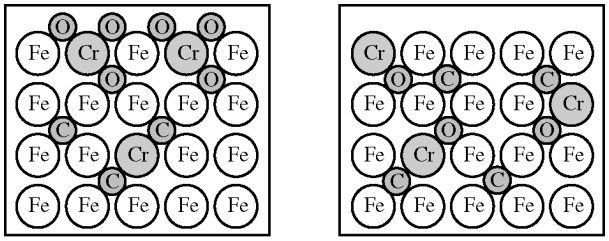
(B) ✓



(C)

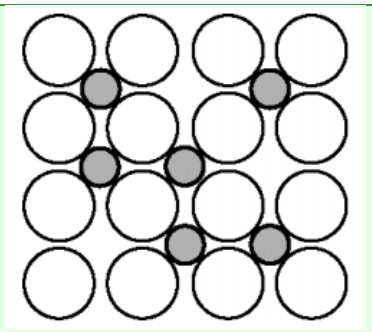


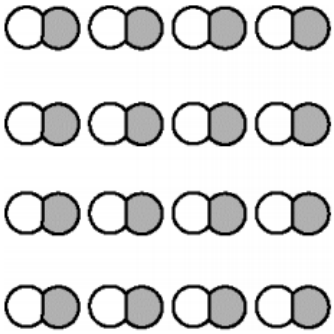
(D)

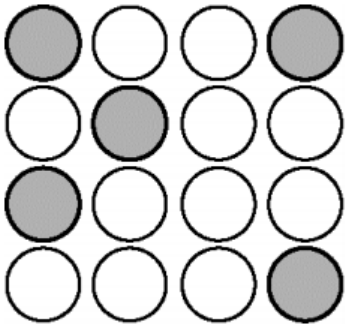


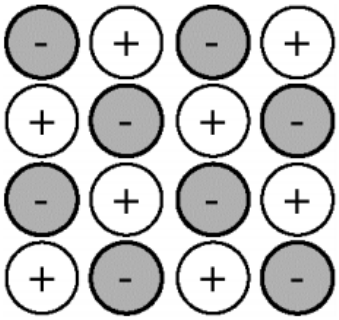
**8.** Which of the following diagrams best depicts an alloy of Ni and B?

**2.4**

(A) 

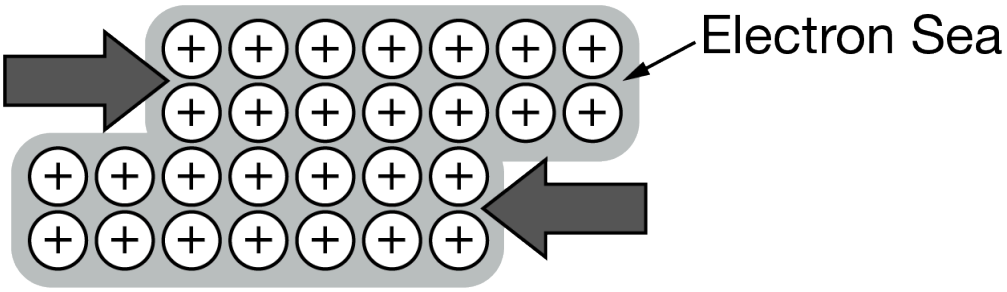
(B) 

(C) 

(D) 

**2.4**

**9.**




The diagram above best illustrates which of the following phenomena associated with solids that have metallic bonding?

(A) Electrical conductivity, because it shows a lattice of positive ions immersed in a sea of electrons.

(B) Malleability, because it shows how adjacent layers of positive ions can move relative to one another while remaining in full contact with the electron sea. ✓

(C) Heat conductivity, because it shows how layers of atoms can slide past one another, creating friction between layers and causing the temperature of the solid to increase.

(D) The ability to form substitutional alloys, because it shows how atoms of two different metals can combine in a one-to-one ratio.

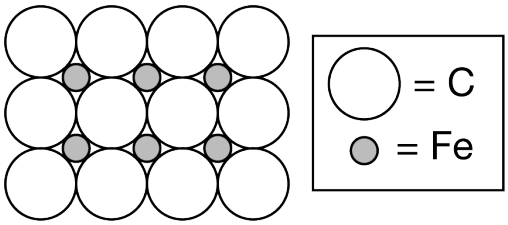
**Answer B**

Correct. The diagram shows the structure of a metallic solid, which has adjacent layers of positive ions sliding past one another (as indicated by the arrows) while staying within the shading of the electron sea.

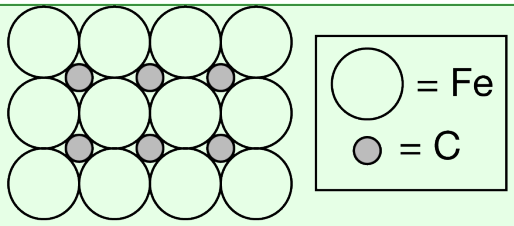
**10.** Steel is an alloy containing $Fe$ atoms and $C$ atoms. Which of the following diagrams best represents the particle-level structure of steel?

**2.4**

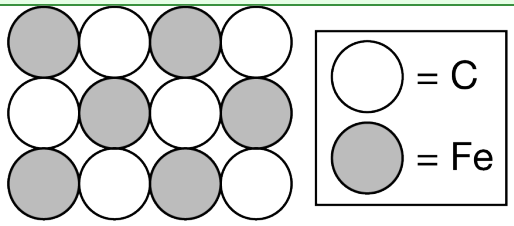
(A)



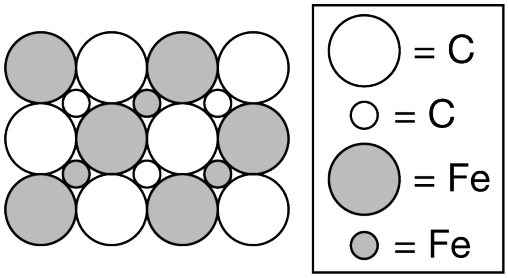

(B) ✓




(C)




(D)




**Answer B**

Correct. C is in the second period and has two electron shells, whereas Fe is in the fourth period and has four electron shells; thus C atoms are smaller than Fe atoms. The diagram shows how the smaller C atoms occupy the spaces (interstices) between the larger Fe atoms in an interstitial alloy.

**11.**

| Type of Steel | % Carbon | Characteristics | Uses |
|---|---|---|---|
| Low-carbon steel | < 0.2 % | Malleable and ductile | Chains and nails |
| High-carbon steel | 0.6 – 1.5 % | Hard and brittle | Cutting tools |

The table above provides some information about two types of steel, both of which are alloys of iron and carbon. Which of the following best helps to explain why high-carbon steel is more rigid than low-carbon steel?

**2.4**

(A) Elemental carbon is harder than elemental iron.

(B) The additional carbon atoms within the alloy make the high-carbon steel less dense.

(C) The additional carbon atoms within the alloy increase the thermal conductivity of the high-carbon steel.

(D) The additional carbon atoms within the alloy make it more difficult for the iron atoms to slide past one another. ✓

## 2.5

**Directions**: Each set of lettered choices below refers to the numbered statements immediately following it. Select the one lettered choice that best fits the statement. A choice may be used once, more than once, or not at all.

The question refers to the following species.

(A) $H_2O$
(B) $NH_3$
(C) $BH_3$
(D) $CH_4$

**1.** Has a central atom with less than an octet of electrons

(A) $H_2O$
(B) $NH_3$
(C) $BH_3$ ✓
(D) $CH_4$

**2.** Has two lone pairs of electrons

(A) $H_2O$ ✓
(B) $NH_3$
(C) $BH_3$
(D) $CH_4$

**Directions**: Each set of lettered choices below refers to the numbered statements immediately following it. Select the one lettered choice that best fits each statement. A choice may be used once, more than once, or not at all in each set.

Refer to the following compounds.

(A) $CH_3CH_2CH_2CH_3$
(B) $CH_3CH_2CH_2OH$
(C) $CH_3COCH_3$
(D) $CH_3COOH$

**3.** Is isomeric with $CH_3CH_2CHO$

(A) $CH_3CH_2CH_2CH_3$
(B) $CH_3CH_2CH_2OH$
(C) $CH_3COCH_3$ ✓
(D) $CH_3COOH$

## 2.5

*Directions*: Each set of lettered choices below refers to the numbered statements immediately following it. Select the one lettered choice that best fits each statement. A choice may be used once, more than once, or not at all in each set.

The following questions refer to the below molecules:

(A) $CO_2$
(B) $H_2O$
(C) $CH_4$
(D) $C_2H_4$
(E) $PH_3$

**4.** The molecule with only one double bond

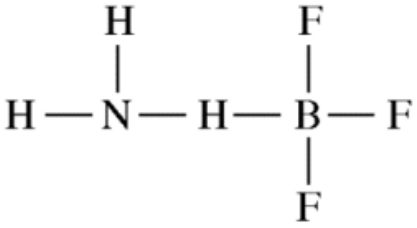
(A) $CO_2$
(B) $H_2O$
(C) $C_2H_4$ ✓
(D) $PH_3$

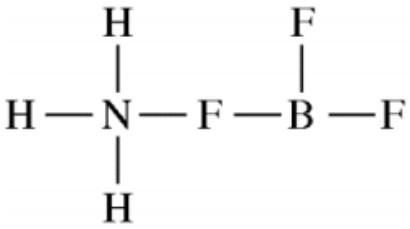
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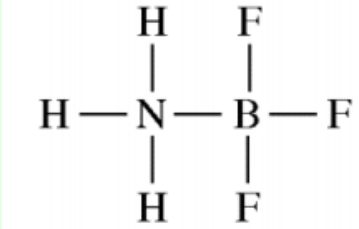
**5.** $NH_3$ reacts with $BF_3$ to form a single species. Which of the following structural diagrams is the most likely representation of the product of the reaction?

**2.5**

(A) 

(B) 

(C) 

(D)  ✓

**6.** The electron-dot structure (Lewis structure) for which of the following molecules would have two unshared pairs of electrons on the central atom?

(A) $H_2S$ ✓

(B) $NH_3$

(C) HCN

(D) $CO_2$

**7.** Which of the following molecules contains only single bonds?

(A) $CH_3COOH$

(B) $C_2H_6$ ✓

(C) $C_6H_6$

(D) HCN

**8.** Which of the following is an isomer of $CH_3OCH_3$ ?

**2.5**

(A) $CH_3CH_3$

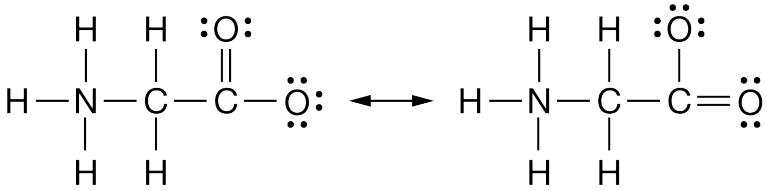
(B) $CH_3COOH$

(C) $CH_3CH_2OH$ ✓

(D) $CH_3CH_2OCH_2CH_3$

## 2.6

**1.**



Based on the resonance structures shown above, what are the bond orders of the two carbon-oxygen bonds?

(A) 1 and 1.5

(B) 1 and 2

(C) 1.5 and 1.5 ✓

(D) 2 and 2

**Answer C**

Correct. When two resonance structures are equivalent, they contribute equally to the average structure, and the average structure is the preferred representation of the molecule. Bond orders are obtained by taking the average of the bond orders in the resonance structures. Therefore, in this case both $C-O$ bonds have bond orders equal to the average of 1 and 2, or 1.5.

**2.** For which of the following molecules are resonance structures necessary to describe the bonding satisfactorily?

(A) $SO_2$ ✓

(B) $CO_2$

(C) $OF_2$

(D) $PF_3$

---

**Directions**: Each set of lettered choices below refers to the numbered statement immediately following it. Select the one lettered choice that best fits the statement. A choice may be used once, more than once, or not at all.

Refer to the following gaseous molecules:

(A) $BeCl_2$

(B) $SO_2$

(C) $N_2$

(D) $F_2$

**2.6**

**3.** Is best represented by two or more resonance forms

(A) $BeCl_2$

(B) $SO_2$ ✓

(C) $N_2$

(D) $F_2$

---

A solution is prepared by combining $1.0\ \text{L}$ of $0.50\ M\ \text{HClO}_2$ with $1.0\ \text{L}$ of $0.50\ M\ \text{KClO}_2$. The $\text{pH}$ of the solution is $2.0$.

**4.** Consider the Lewis electron-dot diagram for the $\text{ClO}_2^-$ ion that minimizes formal charge. Which of the following is true about the hybridization of the $\text{Cl}$ atom in the ion?

(A) The $\text{Cl}$ atom is not hybridized, and the ion contains one single $\text{Cl}-\text{O}$ bond and one double $\text{Cl}-\text{O}$ bond.

(B) The $\text{Cl}$ atom is $sp$ hybridized, and the ion contains two equivalent $\text{Cl}-\text{O}$ double bonds.

(C) The $\text{Cl}$ atom is $sp^2$ hybridized, and the ion contains two equivalent $\text{Cl}-\text{O}$ single bonds.

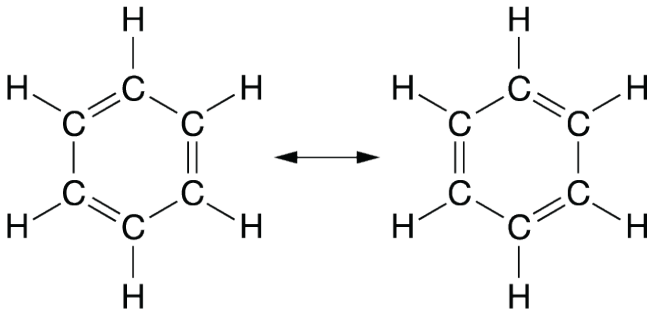
(D) The $\text{Cl}$ atom is $sp^3$ hybridized, and the ion contains two equivalent $\text{Cl}-\text{O}$ single bonds. ✓

**Answer D**

Correct. A valid Lewis electron-dot diagram of the $\text{ClO}_2^-$ ion has two nonbonding pairs of electrons around the central $\text{Cl}$ atom. Since there are four electron domains around the $\text{Cl}$ atom, it is $sp^3$ hybridized.

---

**5.**



The diagram above shows two resonance structures for a molecule of $\text{C}_6\text{H}_6$. The phenomenon shown in the diagram best supports which of the following claims about the bonding in $\text{C}_6\text{H}_6$ ?

---

**2.6**

(A) In the $C_6H_6$ molecule, all the bonds between the carbon atoms have the same length. ✓

(B) Because of variable bonding between its carbon atoms, $C_6H_6$ is a good conductor of electricity.

(C) The bonds between carbon atoms in $C_6H_6$ are unstable, and the compound decomposes quickly.

(D) The $C_6H_6$ molecule contains three single bonds between carbon atoms and three double bonds between carbon atoms.

**Answer A**

Correct. Because of resonance, all the bonds between carbon atoms in $C_6H_6$ are equivalent and have the same strength and equal lengths.

**6.**

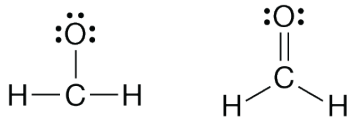


Diagram 1 Diagram 2

Which of the diagrams above best represents the $CH_2O$ molecule, and why?

(A) Diagram 1, because all bond angles are $180^\circ$.

(B) Diagram 1, because all atoms have a formal charge of 0.

(C) Diagram 2, because the molecule has a trigonal pyramidal shape.

(D) Diagram 2, because all atoms have a formal charge of 0. ✓

**Answer D**

Correct. Diagram 2 represents a molecule in which both $C$ and $O$ have a complete octet and all atoms have a formal charge of zero.

**2.6**

**7.**

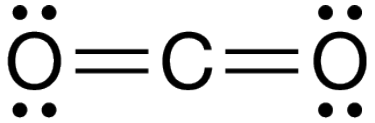


Diagram 1

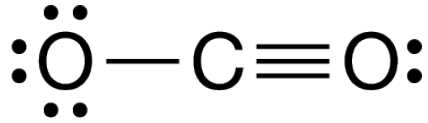


Diagram 2

Which of the Lewis diagrams shown above is the more likely structure of $CO_2$, and why?

(A) Diagram 1, because all the atoms have a formal charge of $0$. ✓

(B) Diagram 1, because double bonds are stronger than triple bonds.

(C) Diagram 2, because all the atoms have a formal charge of $0$.

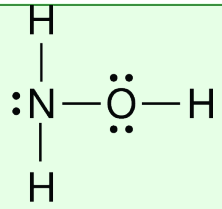
(D) Diagram 2, because triple bonds are stronger than double bonds.

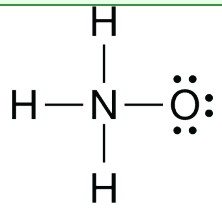
**Answer A**

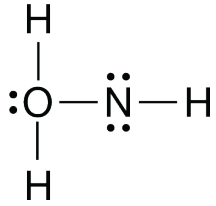
Correct. When the formal charge of each atom in diagram 1 is determined using the rule $\left(\text{formal charge of an atom} = \text{the number of valence electrons of the atom} - \left(\text{the number of electrons in lone pairs} + \text{half the number of electrons in bonds with other atoms}\right)\right)$, the results are zero for each of the three atoms. A molecular structure in which all formal charges are zero is preferable to one in which some formal charges are not zero. And in diagram 2, the results for the three atoms (going left to right) are $-1$, $0$, and $+1$, respectively.

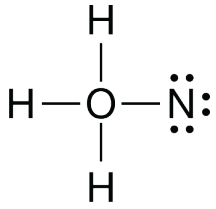
**8.** Based on formal charges, which of the following is the best Lewis electron-dot diagram for $H_3NO$?

**2.6**

(A)  ✓

(B) 

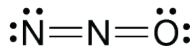
(C) 

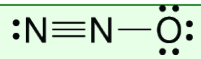
(D) 

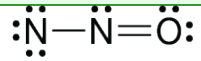
**Answer A**

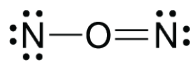
Correct. When the formal charge of each atom in this diagram is determined using the rule (formal charge of an atom = the number of valence electrons of the atom − (the number of electrons in lone pairs + half the number of electrons in bonds with other atoms)), the sum of the formal charges on all the atoms in $H_3NO$ is equal to zero. A sum of formal charges on all atoms equal to zero is preferable.

**9.** Which of the following Lewis diagrams best represents the bonding in the $N_2O$ molecule, considering formal charges?

(A) 

(B)  ✓

(C) 

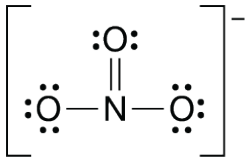
(D) 

**2.6**

**Answer B**

Correct. When the formal charge of each atom in this diagram is determined using the rule: $(\text{formal charge of an atom} = \text{the number of valence electrons of the atom} - (\text{the number of electrons in lone pairs} + \text{half the number of electrons in bonds with other atoms}))$. The sum of the formal charges on all the atoms in $N_2O$ is equal to zero. The most favored structure considers the minimization of the formal charges and the electronegativities of the atoms. Oxygen is more electronegative than nitrogen; therefore, the Lewis diagram that shows a negative formal charge on the oxygen atom is the best for depicting the molecule.

**10.**



Which of the following statements, if true, would support the claim that the $NO_3^-$ ion, represented above, has three resonance structures?

(A) The $NO_3^-$ ion is not a polar species. ✓

(B) The oxygen-to-nitrogen-to-oxygen bond angles are $90^\circ$.

(C) One of the bonds in $NO_3^-$ is longer than the other two.

(D) One of the bonds in $NO_3^-$ is shorter than the other two.

**Answer A**

Correct. If the $NO_3^-$ ion did NOT have three equivalent structures, then the ion would be polar because of the unequal distribution of charge within the ion. Equal contributions of the three resonance forms to the structure of the ion result in a planar ion with three equivalent polar nitrogen-oxygen bonds. Due to the trigonal planar symmetry of the $NO_3^-$ ion, the negative dipoles cancel out.

**11.** Which of the following molecules has an angular (bent) geometry that is commonly represented as a resonance hybrid of two or more electron-dot structures?

(A) $CO_2$

(B) $O_3$ ✓

(C) $BeF_2$

(D) $OF_2$

**2.6**

**12.** Resonance is most commonly used to describe the bonding in molecules of which of the following?

(A) $CO_2$

(B) $O_3$ ✓

(C) $H_2O$

(D) $SF_6$

## 2.7

**1.** According to the VSEPR model, the progressive decrease in the bond angles in the series of molecules $CH_4$, $NH_3$, and $H_2O$ is best accounted for by the

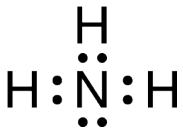
(A) increasing strength of the bonds

(B) increasing electronegativity of the central atom

(C) increasing number of unshared pairs of electrons ✓

(D) decreasing repulsion between hydrogen atoms

**2.**



Based on the Lewis diagram for $NH_3$, shown above, the H-N-H bond angle is closest to which of the following?

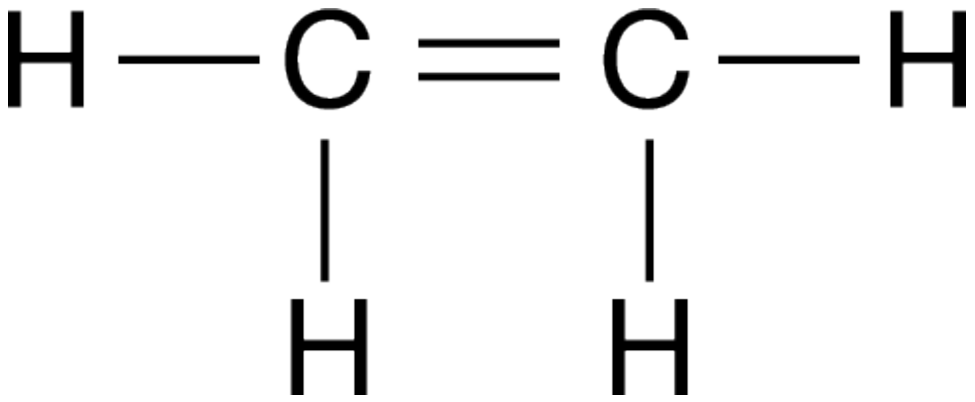
(A) $60^\circ$

(B) $90^\circ$

(C) $109.5^\circ$ ✓

(D) $120^\circ$

**3.**



A Lewis diagram for the molecule $C_2H_4$ is shown above. In the actual $C_2H_4$ molecule, the H-C-H bond angles are closest to

(A) $90^\circ$

(B) $109.5^\circ$

(C) $120^\circ$ ✓

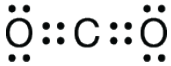
(D) $180^\circ$

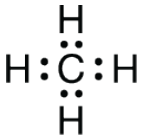
**2.7**

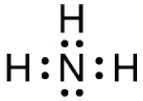
**Answer C**

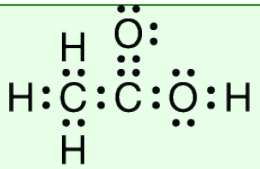
Correct. With three electron domains around each C atom, their geometry is trigonal planar (which minimizes repulsion among the three electron pairs). Therefore the H-C-H bond angle is approximately $120^\circ$.

**4.** Which of the following complete Lewis diagrams represents a molecule containing a bond angle that is closest to $120^\circ$?

(A) 

(B) 

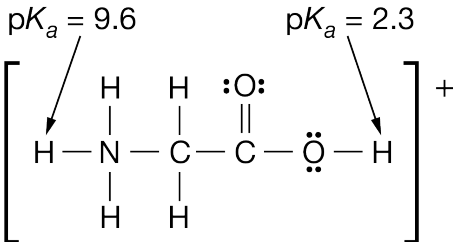
(C) 

(D) 

**Answer D**

Correct. The carbon atom attached to the oxygen atoms exhibits $sp^2$ hybridization, thus the bond angles around it are close to $120^\circ$. Another explanation is, the carbon atom attached to the two oxygen atoms is surrounded by three electron domains, which, according to VSEPR theory, will repel one another to produce a trigonal planar geometry around that atom.

**2.7**




The structural formula of the glycinium cation is shown above. Arrows indicate the $\mathrm{p}K_a$ values for the labile protons in the molecule.

**5.** What is the approximate $\mathrm{H} - \mathrm{O} - \mathrm{C}$ bond angle in the glycinium cation?

(A) $180^\circ$

(B) $120^\circ$

(C) $105^\circ$ ✓

(D) $90^\circ$

**Answer C**

Correct. The oxygen atom in the $\mathrm{H} - \mathrm{O} - \mathrm{C}$ bond has two lone pairs of electrons, so the geometry of the electron domains around the oxygen atom is roughly tetrahedral. The bond angle is thus close to $109^\circ$.

**6.** Which of the following is true about the geometry of the glycinium cation?

(A) The leftmost $\mathrm{C}$ atom and all the atoms directly bonded to it lie in the same plane.

(B) Both $\mathrm{C}$ atoms and both $\mathrm{O}$ atoms lie in the same plane. ✓

(C) The $\mathrm{N} - \mathrm{C} - \mathrm{C}$ bond angle is $180^\circ$.

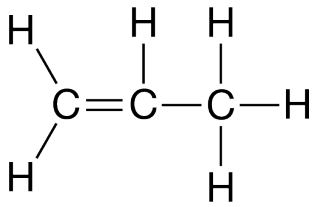
(D) The geometry around the $\mathrm{N}$ atom is planar.

**Answer B**

Correct. The carbon atom bonded to two oxygen atoms and another carbon atom does not have any nonbonding pairs of electrons, thus the central carbon atom has three electron domains. According to $\mathrm{VSEPR}$ theory, all four atoms are in the same plane ($sp^2$ hybridization).

**2.7**

**7.**



Which of the following correctly compares the strength of the two carbon-to-carbon bonds in the molecule represented in the Lewis diagram shown above?

(A) The carbon-to-carbon bond on the left is stronger because it is a double bond. ✓

(B) The carbon-to-carbon bonds are the same strength because the C-C-C bond angle is $180^\circ$.

(C) The carbon-to-carbon bonds are the same strength because they are both bonds between carbon atoms.

(D) The carbon-to-carbon bond on the right is stronger because there are more hydrogen atoms attached to the rightmost carbon atom than to the leftmost carbon atom.

**Answer A**

Correct. Double bonds between two atoms are always stronger than single bonds between those same two atoms because in a double bond there are more electrons between the two positive nuclei of the atoms, which results in a stronger bond.

---

Reaction 1: $CaC_2(s) + 2\ H_2O(l) \rightarrow C_2H_2(g) + Ca(OH)_2(s)$

Reaction 2: $NaOCl(aq) + 2\ HCl(aq) \rightarrow Cl_2(g) + NaCl(aq) + H_2O(l)$

Reaction 3: $C_2H_2(g) + Cl_2(g) \rightarrow C_2H_2Cl_2(g)$

**8.** When Reaction 3 occurs, does the hybridization of the carbon atoms change?

(A) Yes; it changes from *sp* to $sp^2$. ✓

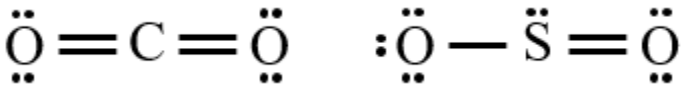
(B) Yes; it changes from *sp* to $sp^3$.

(C) Yes; it changes from $sp^2$ to $sp^3$.

(D) No; it does not change

---

**2.7**

**9.** 

Lewis electron-dot diagrams for $CO_2$ and $SO_2$ are given above. The molecular geometry and polarity of the two substances are

(A) the same because the molecular formulas are similar

(B) the same because C and S have similar electronegativity values

(C) different because the lone pair of electrons on the S atom make it the negative end of a dipole

(D) different because S has a greater number of electron domains (regions of electron density) surrounding it than C has ✓

**Answer D**

This option is correct. Both the C-O and S-O bonds are polar, due to the different electronegativities of the bonded atoms. However, the $CO_2$ molecule is linear and so the bond dipoles cancel and the molecule has no net dipole. In $SO_2$, the molecule is bent and so the bond dipoles do not cancel and a net dipole is present. The polarity of the two substances differs because of the differences in geometry, which is a direct consequence of the number of electron domains around the central atom (2 in $CO_2$ and 3 in $SO_2$), as stated in this option.

---

Refer to the following diatomic species

(A) $Li_2$

(B) $B_2$

(C) $N_2$

(D) $O_2$

(E) $F_2$

**10.** Contains 1 sigma (σ) and 2 pi (π) bonds

(A) $B_2$

(B) $N_2$ ✓

(C) $O_2$

(D) $F_2$

**2.7**

**11.** Has a bond order of 2

(A) $B_2$

(B) $N_2$

(C) $O_2$ ✓

(D) $F_2$

**12.** Has the largest bond-dissociation energy

(A) $B_2$

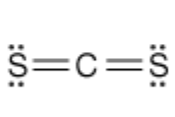
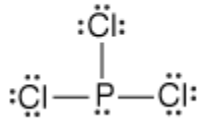
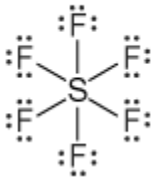
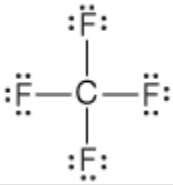
(B) $N_2$ ✓

(C) $O_2$

(D) $F_2$

---

**13.**

| $CS_2$ | $PCl_3$ | $SF_6$ | $CF_4$ |
|---|---|---|---|
|  |  |  |  |

Based on the diagrams shown above, which of the following would have the greatest dipole moment, and why?

(A) $CS_2$, because the double bonds create an area of higher electron density around the C atom.

(B) $PCl_3$, because the Cl atoms draw electron density away from the P atom and the bond dipoles do not cancel one another. ✓

(C) $SF_6$, because it has the greatest number of polar bonds.

(D) $CF_4$, because the large electronegativity difference between C and F results in very polar bonds.

**Answer B**

Correct. $PCl_3$ is predicted by Valence Shell Electron Pair Repulsion (VSEPR) theory to have a trigonal pyramidal molecular shape. The central P atom has four electron domains around it, giving a tetrahedral electronic geometry with one of the positions occupied by a lone pair of electrons. The bond dipoles would not cancel, and $PCl_3$ would have a nonzero dipole moment.

**2.7**

---

**Directions**: Each set of lettered choices below refers to the numbered statement immediately following it. Select the one lettered choice that best fits the statement. A choice may be used once, more than once, or not at all.

Refer to the following gaseous molecules:

(A) $BeCl_2$

(B) $SO_2$

(C) $N_2$

(D) $F_2$

**14.** Is the molecule in which the intramolecular forces are strongest

(A) $BeCl_2$
(B) $SO_2$
(C) $N_2$ ✓
(D) $F_2$

**15.** Is a polar molecule

(A) $BeCl_2$
(B) $SO_2$ ✓
(C) $N_2$
(D) $F_2$

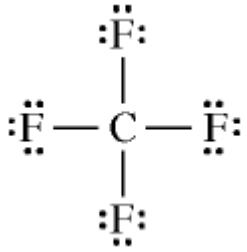
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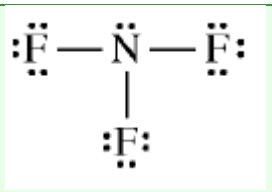
**16.** The geometry of the $SO_3$ molecule is best described as

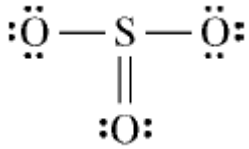
(A) trigonal planar ✓
(B) trigonal pyramidal
(C) bent
(D) tetrahedral

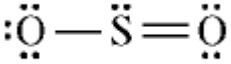
**17.** Which of the following Lewis electron-dot diagrams represents the molecule that contains the smallest bond angle?

**2.7**

(A) 

(B) 

(C) 

(D) 

---

**Directions**: Each set of lettered choices below refers to the numbered statements immediately following it. Select the one lettered choice that best fits the statement. A choice may be used once, more than once, or not at all.

The question refers to the following species.

(A) $H_2O$
(B) $NH_3$
(C) $BH_3$
(D) $CH_4$

**18.** Has a trigonal-pyramidal molecular geometry

(A) $H_2O$
(B) $NH_3$ ✓
(C) $BH_3$
(D) $CH_4$

**19.** Is predicted to have the largest bond angle

**2.7**

(A) $H_2O$
(B) $NH_3$
(C) $BH_3$ ✓
(D) $CH_4$

---

**20.** Which of the following has molecules with a pyramidal shape?

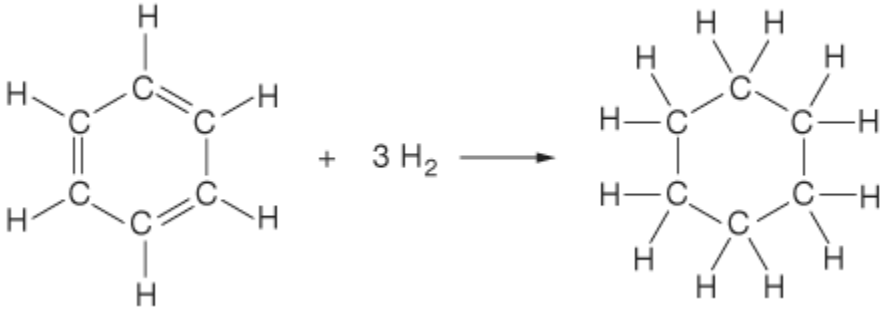
(A) $NH_3(g)$ ✓
(B) $BH_3(g)$
(C) $H_2S(g)$
(D) HBr(g)

**21.** H - C ≡ C - H

What is the hybridization of the carbon atoms in a molecule of ethyne, represented above?

(A) *sp* ✓
(B) $sp^2$
(C) $sp^3$
(D) $dsp^2$

**22.**




In the reaction represented above, which of the following sets correctly identifies the hybridization of the C atoms before and after the reaction occurs?

| Set | Before | After |
|---|---|---|
| **A** | $sp^2$ | $sp^2$ |
| **B** | $sp^2$ | $sp^3$ |
| **C** | $sp^3$ | $sp^2$ |
| **D** | $sp^3$ | $sp^3$ |

**2.7**

(A) Set A

(B) Set B ✓

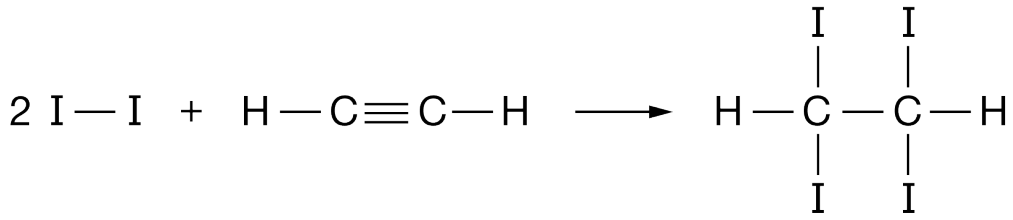
(C) Set C

(D) Set D

**Answer B**

Correct. In the $C_6H_6$ molecule, each carbon atom is surrounded by three electron domains in a trigonal planar geometry, and the hybridization of the orbitals of each carbon atom is $sp^2$. In the $C_6H_{12}$ molecule, each carbon atom is surrounded by four electron domains in a tetrahedral geometry, and the hybridization of the orbitals of each carbon atom is $sp^3$.

**23.** The reaction between $I_2$ and $C_2H_4$ is represented by the following equation.



How does the hybridization of the carbon atoms change from reactants to products?

(A) $sp$ to $sp^2$

(B) $sp$ to $sp^3$ ✓

(C) $sp^2$ to $sp$

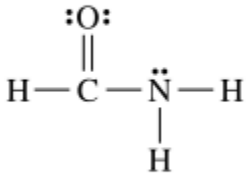
(D) $sp^3$ to $sp$

**Answer B**

Correct. Each C atom in $C_2H_4$ has two electron domains, so they have $sp$ hybridization. Each C atom in $C_2H_2I_4$ has four electron domains, so they have $sp^3$ hybridization. The hybridization of the carbon atoms therefore changes from $sp$ to $sp^3$.

**2.7**

**24.**



The complete Lewis electron-dot diagram for $HCONH_2$ is shown above. Identify the hybridization of the valence orbitals for the carbon atom and the approximate hydrogen-carbon-nitrogen bond angle according to VSEPR theory.

(A) $sp$ hybridized and $180^\circ$

(B) $sp^2$ hybridized and $120^\circ$ ✓

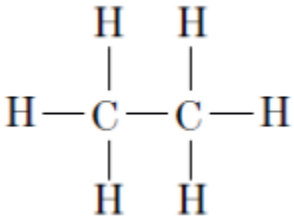
(C) $sp^2$ hybridized and $180^\circ$

(D) $sp^3$ hybridized and $120^\circ$

**Answer B**

Correct. The carbon atom has three electron domains: the $C-H$ bond, the $C-N$ bond, and the $C=O$ bond. According to VSEPR theory, three domains around an atom achieve maximum separation and greatest stability if the domains are arranged in a trigonal planar fashion, with $120^\circ$ bond angles. When there are three electron domains, the valence orbitals are $sp^2$ hybridized.

**25.**

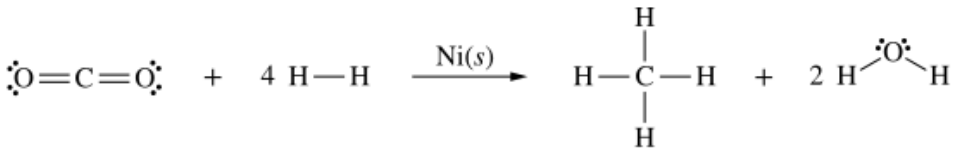


The hybridization of the carbon atoms in the molecule represented above can be described as

**2.7**

(A) *sp*
(B) $sp^2$
(C) $sp^3$ ✓
(D) $dsp^2$

In the gas-phase reaction represented by the following equation, methane $(CH_4)$ and water $(H_2O)$ are produced from carbon dioxide $(CO_2)$ and hydrogen $(H_2)$ at high temperature and pressure.




**26.** Which of the following identifies the hybridization of the valence orbitals of the central atoms of the methane and water molecules?

| | $CH_4$ | $H_2O$ |
|---|---|---|
| A | $sp^2$ | $sp$ |
| B | $sp^2$ | $sp^2$ |
| C | $sp^3$ | $sp^2$ |
| D | $sp^3$ | $sp^3$ |

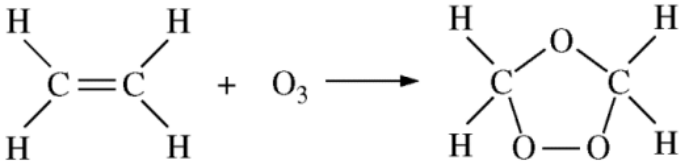
(A) A
(B) B
(C) C
(D) D ✓

**Answer D**

Correct. The hybridization of the valence orbitals of the central atoms of $CH_4$ and $H_2O$ are both $sp^3$ because both have four electron domains around the central atom.

**2.7**

**27.**



In the reaction represented above, what is the hybridization of the C atoms before and after the reaction occurs?

(A)

| Before | After |
|---|---|
| *sp* | $sp^2$ |

(B)

| Before | After |
|---|---|
| sp | $sp^3$ |

(C)

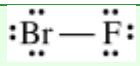
| Before | After |
|---|---|
| $sp^2$ | sp |

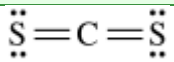
(D) ✓

| Before | After |
|---|---|
| $sp^2$ | $sp^3$ |

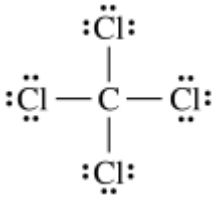
**28.** Which of the following molecules has the largest dipole moment?

**2.7**

(A)  ✓

(B) 

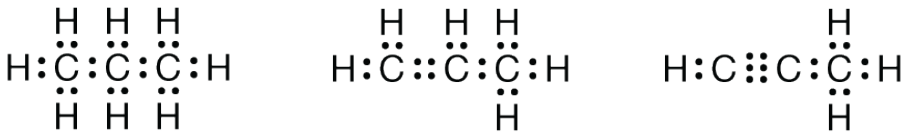
(C) 

(D) 

**Answer A**

Correct. The molecule with the largest dipole moment has bond dipoles that do not cancel as a result of the molecular shape. The BrF molecule has a bond dipole and the molecule is not symmetrical, so the bond dipoles do not cancel.

**29.**



Molecule 1 Molecule 2 Molecule 3

Lewis diagrams of molecules of three different hydrocarbons are shown above. Which of the following claims about the molecules is best supported by the diagrams?

(A) All the atoms in molecule 1 lie in one plane.

(B) All the molecules have the same empirical formula.

(C) The C-C-C bond angle in molecule 2 is close to $180^\circ$.

(D) The strongest carbon-to-carbon bond occurs in molecule 3. ✓

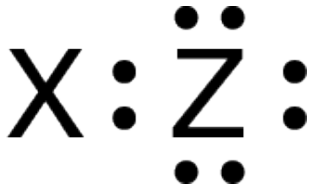
**Answer D**

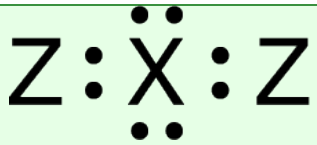
Correct. Molecule 3 contains a triple bond between two carbon atoms. Between two of the same kind of

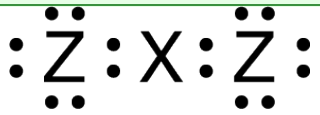
**2.7**

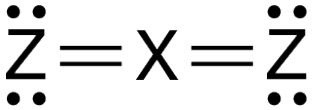
atom, triple bonds are stronger than double bonds, which in turn are stronger than single bonds.

**30.** In the following diagrams, elements are represented by X and Z, which form molecular compounds with one another. Which diagram represents a molecule that has a bent molecular geometry?

(A) 

(B)  ✓

(C) 

(D) 

**Answer B**

Correct. There are four electron domains around the central atom X, and their geometry is tetrahedral, which minimizes the repulsion between the four electron pairs. The Z-X-Z bond angle is approximately $109.5^\circ$, and the molecular geometry is bent.

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*Directions*: Each set of lettered choices below refers to the numbered statements immediately following it. Select the one lettered choice that best fits each statement. A choice may be used once, more than once, or not at all in each set.

The following questions refer to the below molecules:

(A) $CO_2$
(B) $H_2O$
(C) $CH_4$
(D) $C_2H_4$
(E) $PH_3$

**31.** The molecule with the largest dipole moment

**2.7**

(A) $CO_2$

(B) $H_2O$ ✓

(C) $C_2H_4$

(D) $PH_3$

**32.** The molecule that has trigonal pyramidal geometry

(A) $CO_2$

(B) $H_2O$

(C) $C_2H_4$

(D) $PH_3$ ✓

---

**33.** Which of the following molecules contains exactly three sigma (σ) bonds and two pi (π) bonds?

(A) $C_2H_2$ ✓

(B) $CO_2$

(C) HCN

(D) $SO_3$

**34.** Of the following molecules, which has the largest dipole moment?

(A) CO

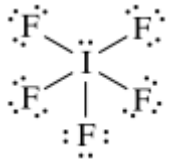
(B) $CO_2$

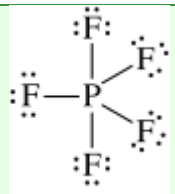
(C) HF ✓

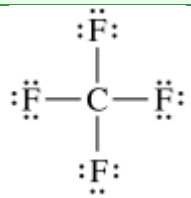
(D) $F_2$

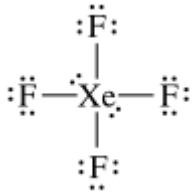
**35.** Which of the following represents a molecule with a trigonal bipyramidal geometry?

**2.7**

(A) 

(B)  ✓

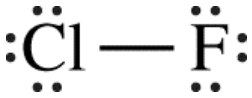
(C) 

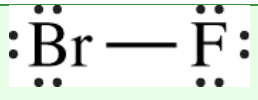
(D) 

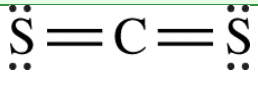
**Answer B**

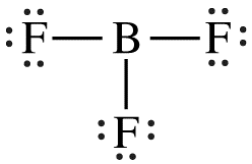
Correct. The Lewis diagram for a molecule having trigonal bipyramidal geometry must have five bonding pairs and no nonbonding pairs of electrons around the central atom. This response has the required arrangement of bonding and nonbonding pairs around the central P atom.

**36.** Which of the following Lewis electron-dot diagrams represents the molecule that is the most polar?

(A) 

(B)  ✓

(C) 

(D) 

**2.7**

**37.** Which of the following molecules is nonpolar but has polar covalent bonds?

(A) $H_2O_2$

(B) $H_2O$

(C) $CCl_4$ ✓

(D) $CH_2Cl_2$

**38.** Which of the following arranges the molecules $N_2$, $O_2$, and $F_2$ in order of their bond enthalpies, from least to greatest?

(A) $F_2 < O_2 < N_2$ ✓

(B) $O_2 < N_2 < F_2$

(C) $N_2 < O_2 < F_2$

(D) $N_2 < F_2 < O_2$

**39.** Pi (π) bonding occurs in each of the following species EXCEPT

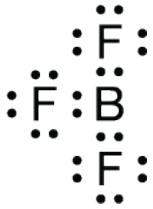
(A) $CO_2$

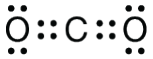
(B) $CN^-$

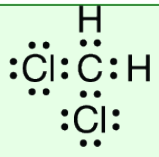
(C) $C_6H_6$

(D) $CH_4$ ✓

**40.** Which of the following Lewis diagrams represents a molecule that is polar?

(A) 

(B) 

(C)  ✓

(D) 

**Answer C**

Correct. In the $CH_2Cl_2$ molecule, the H and Cl atoms are in a tetrahedral arrangement around the

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central C atom. The two electronegative chlorine atoms create a partial negative charge on their side of the carbon atom, leaving a partial positive charge on the opposite (hydrogen) side of the carbon atom. This separation of charge constitutes a dipole, and the molecule is polar.

---

An equimolar mixture of $SO_2$ and $O_2$ is placed in a rigid container at $298\ K$. The initial pressure in the container is $0.20\ atm$. A catalyst is added to the mixture, and the reaction represented below occurs.

$$2\ SO_2(g) + O_2(g) \rightarrow 2\ SO_3(g) \qquad \Delta H^\circ = -200\ kJ/mol_{rxn}$$

**41.** Which of the following gives the shape of the $SO_2$ molecule and best helps explain why it has that shape?

(A) It is linear, because there are two electron domains around the sulfur atom.

(B) It is linear, because there are three electron domains around the sulfur atom.

(C) It is bent, because there are two electron domains around the sulfur atom.

(D) It is bent, because there are three electron domains around the sulfur atom. ✓

**Answer D**

Correct. A Lewis electron-dot diagram for $SO_2$ can be drawn with two equivalent resonance structures. S is the central atom, and there is one single bond and one double bond, with resonance between the two bonds. The S atom has one nonbonding pair of electrons. Since there are two bonds and one nonbonding pair of electrons on the S atom, the molecule is bent.

---

**42.** The $BF_3$ molecule is nonpolar, whereas the $NF_3$ molecule is polar. Which of the following statements accounts for the difference in polarity of the two molecules?

(A) In $NF_3$, each F is joined to N with multiple bonds, whereas in $BF_3$ , each F is joined to B with single bonds.

(B) N — F bonds are polar, whereas B — F bonds are nonpolar.

(C) $NF_3$ is an ionic compound, whereas $BF_3$ is a molecular compound.

(D) Unlike $BF_3$, $NF_3$ has a nonplanar geometry due to an unshared pair of electrons on the N atom. ✓

**43.** Which of the molecules represented below contains carbon with $sp^2$ hybridization?

**2.7**

(A) $CH_2Cl_2$

(B) $C_2H_6$

(C) $C_2H_2Cl_2$ ✓

(D) $C_2H_4Cl_2$

**44.** Which of the following has a zero dipole moment?

(A) HCN

(B) $SO_2$

(C) $NO_2$

(D) $PF_5$ ✓

**45.** Which of the following species is NOT planar?

(A) $CO_3^{2-}$

(B) $NO_3^-$

(C) $ClF_3$

(D) $PCl_3$ ✓

## Unit 2 Progress Check: FRQ

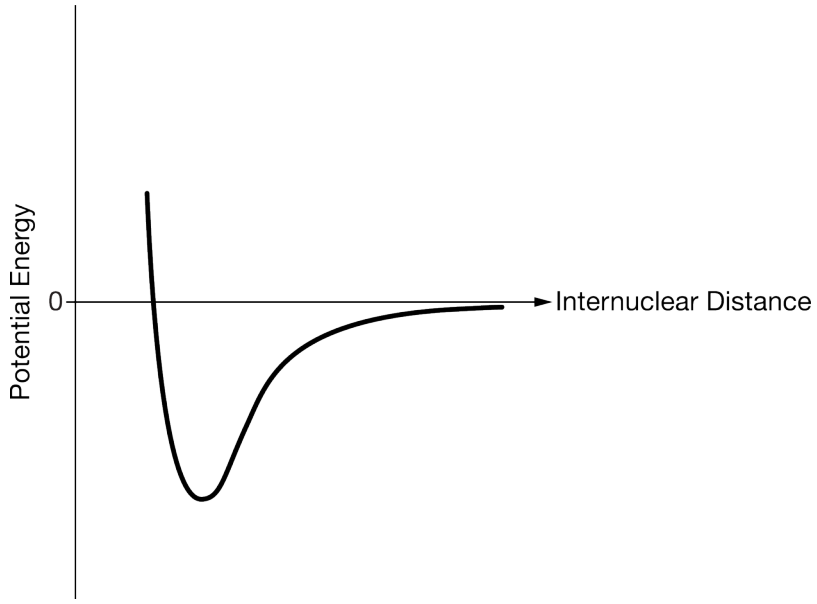
**1.** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Answer the following questions about $N_2$ and $N_2H_4$.

(a) In the box below, draw the complete Lewis electron-dot diagram of $N_2$ .

(b) Based on the Lewis electron-dot diagram that you drew, is the $N_2$ molecule polar? Explain.

The following graph shows the potential energy of two nitrogen atoms versus the distance between their nuclei.




(c) On the graph, indicate the distance that corresponds to the bond length of the $N_2$ molecule by placing an $X$ on the horizontal axis.

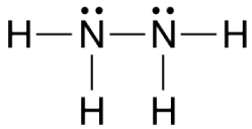
(d) On the graph, which shows the potential energy curve of two $N$ atoms, carefully sketch a curve that corresponds to the potential energy of two $O$ atoms versus the distance between their nuclei.

## Unit 2 Progress Check: FRQ

$N_2$ can react with $H_2$ to form the compound $N_2H_4$.

(e) A sample of $N_2H_4$ has a mass of $25$ g. Identify the numerical quantity that is needed to convert the number of grams of $N_2H_4$ to the number of moles of $N_2H_4$. (You do not need to do the actual calculation.)

(f) The Lewis electron-dot diagram of $N_2H_4$ is shown below.



(i) Based on the Lewis electron-dot diagrams of $N_2$ and $N_2H_4$, compare the length of the nitrogen-to-nitrogen bond in $N_2$ with the length of the nitrogen-to-nitrogen bond in $N_2H_4$.

(ii) Based on the Lewis electron-dot diagrams of $N_2$ and $N_2H_4$, compare the strength of the nitrogen-to-nitrogen bond in $N_2$ with the strength of the nitrogen-to-nitrogen bond in $N_2H_4$.

(iii) Identify the hybridization of the $N$ atoms in $N_2H_4$.

**Part (a)**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 | 2 ✓ |
|---|---|---|

The response meets both of the criteria below:

- ☐ The diagram shows a total of ten electrons.
- ☐ The diagram shows three pairs of electrons between the $N$ two atoms.

**Part (b)**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 ✓ |
|---|---|

The response indicates that the $N_2$molecule is not polar, because the three bonding pairs of electrons are shared equally between the two $N$ atoms (they have the same electronegativity), and each $N$ atom has a nonbonding electron pair (the molecule is completely symmetrical).

**Part (c)**

## Unit 2 Progress Check: FRQ

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 ✓ |
|---|---|

The response shows an $\mathrm{X}$ drawn on the axis and located directly above the lowest part (minimum) of the potential energy curve. The lowest potential energy on the curve corresponds to the most stable internuclear distance (i.e., bond length).

**Part (d)**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 | 2 ✓ |
|---|---|---|

The sketched curve meets both of the criteria below:

- ☐ The curve has its minimum to the right of where the given curve has its minimum (because the length of the $\mathrm{O}$-$\mathrm{O}$ double bond is greater than that of the $\mathrm{N}$-$\mathrm{N}$ triple bond).
- ☐ The minimum value of the curve is higher than the minimum of the given curve (because the energy of the $\mathrm{O}$-$\mathrm{O}$ double bond is lower than that of the $\mathrm{N}$-$\mathrm{N}$ triple bond).

**Part (e)**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 ✓ |
|---|---|

The response indicates that the molar mass of $\mathrm{N_2H_4}$ (or $32\ \mathrm{g/mol}$ ) is needed to solve the problem.

**Part (f)(i)**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 ✓ |
|---|---|

The response indicates that the nitrogen-to-nitrogen bond in $\mathrm{N_2}$ is shorter than the nitrogen-to-nitrogen bond in $\mathrm{N_2H_4}$ . (No explanation is required, however, the length of a triple bond is shorter than that of a single bond.)

## Unit 2 Progress Check: FRQ

### Part (f)(ii)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 ✓ |
|---|---|

The response indicates that the nitrogen-to-nitrogen bond in $N_2$ is stronger than the nitrogen-to-nitrogen bond in $N_2H_4$ . (No explanation is required, however, the strength of a triple bond is greater than that of a single bond.)

### Part (f)(iii)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

| 0 | 1 ✓ |
|---|---|

The response indicates that the hybridization of the $N$ atoms is $sp^3$ . (No explanation is required, however, the $N$ atoms in $N_2H_4$ are surrounded by four electron domains, one of which is a lone pair. This situation corresponds to $sp^3$ , hybridization.)